

Bayesian Core: Introduction & Foundations

University Notes

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1 Lesson 1: The Bayesian Paradigm

Welcome to the Bayesian side of statistics. As the famous quote by Obi-Wan Kenobi in *Star Wars: Return of the Jedi* states: “Luke, you’re going to find that many of the truths we cling to depend greatly on our own point of view”. This philosophical stance elegantly captures the essence of Bayesian statistics, where probability quantifies a degree of belief or confidence in an event’s truth, rather than merely reflecting long-run frequencies. In this foundational chapter, we will introduce elementary examples of Bayes’ theorem, review core concepts of probability including random variables and densities, and establish the “Bayesian core” concepts: the **prior**, the **likelihood**, and the **posterior**.

1.1 Review of Probability Concepts

To build our Bayesian models, we must first recognize that almost everything is random.

Definition: Axioms of Probability

The probability $P(A)$ of an event $A \subset \Omega$ is a number in the interval $[0, 1]$. The foundational rules state that:

1. $0 \leq P(A) \leq 1$ for any event A
2. $P(\Omega) = 1$, meaning the probability of the entire sample space is certain
3. $P(\cup_n A_n) = \sum_n P(A_n)$ whenever A_n are mutually disjoint events

1.1.1 The Bayesian Toolbox

Before performing complex statistical inference, we must master two essential probabilistic tools: conditional probability and total probability.

Theorem & Rules: Conditional & Total Probability

Conditional Probability: If an event H has a probability $P(H) > 0$, the conditional probability (or posterior) of an event E given H has occurred is defined as:

$$P(E|H) := \frac{P(E \cap H)}{P(H)}$$

Total Probability: If the sample space $\Omega = H_1 \cup H_2$, where $H_1 \cap H_2 = \emptyset$, and both $P(H_1) > 0$ and $P(H_2) > 0$, then the marginal probability of E is:

$$P(E) = P(E|H_1)P(H_1) + P(E|H_2)P(H_2)$$

1.2 Bayes' Theorem: The Engine of Learning

Bayes' theorem essentially represents the inversion of probabilities. It allows us to update our beliefs when new evidence is presented.

Theorem & Rules: Bayes' Theorem

If E , H_1 , and H_2 are events such that $P(H_1)$, $P(H_2)$, $P(E) \neq 0$ and they partition the space ($\Omega = H_1 \cup H_2$, $H_1 \cap H_2 = \emptyset$), then the probability of H_1 given E is:

$$P(H_1|E) = \frac{P(E|H_1)P(H_1)}{P(E|H_1)P(H_1) + P(E|H_2)P(H_2)}$$

To understand how this operates in practice, consider the following practical diagnostic scenario.

Example: Circuit Diagnosis

Imagine a circuit is not working. We formulate two mutually exclusive hypotheses:

- H_1 : The battery is dead.
- H_2 : The battery is working, meaning the failure is due to other causes ($H_2 = H_1^c$).

Step 1: The Prior. Based on previous experience, we assign an initial belief of a 20% chance for a dead battery ($P(H_1) = 0.2$) and an 80% chance for other causes ($P(H_2) = 0.8$).

Step 2: The Evidence & Likelihood. A voltmeter test yields the observation E : “low voltage”. Based on our model, we know:

- The probability of showing low voltage given a dead battery is 0.9 ($P(E|H_1) = 0.9$).
- The probability of showing low voltage given a working battery is 0.1 ($P(E|H_2) = 0.1$).

Step 3: Total Probability. We calculate the overall probability of seeing a low voltage reading:

$$P(E) = (0.9 \cdot 0.2) + (0.1 \cdot 0.8) = 0.26$$

Step 4: The Posterior. Using Bayes' theorem, we compute the updated probability that the battery is dead, given the low voltage reading:

$$P(H_1|E) = \frac{0.9 \cdot 0.2}{0.26} \approx 0.692$$

Important Note

Takeaways on the Learning Process: The posterior, $P(H|E)$, is the updated probability of the hypothesis H given we observed E . The prior probability was 20%, but after conducting the experiment, the posterior probability jumped to 69%. The experiment significantly increases the likelihood of a dead battery, demonstrating how Bayes' Theorem helps us update probabilities with new evidence and make better decisions under uncertainty.

Example: The Urns

Consider two urns, each containing four balls. Urn 1 has 3 black balls and 1 white ball, while Urn 2 has 2 black balls and 2 white balls.

If we randomly pick an urn and then draw one ball, what is the conditional (posterior) probability that the ball came from Urn 1, given that the drawn ball is black?

$$\begin{aligned} P(\text{Urn 1}|\text{Black}) &= \frac{P(\text{Black}|\text{Urn 1})P(\text{Urn 1})}{P(\text{Black}|\text{Urn 1})P(\text{Urn 1}) + P(\text{Black}|\text{Urn 2})P(\text{Urn 2})} \\ &= \frac{(3/4) \cdot (1/2)}{(3/4) \cdot (1/2) + (2/4) \cdot (1/2)} = \frac{3}{5} \end{aligned}$$

1.3 From Discrete Events to Continuous Models

What happens when our data are not simple 0 – 1 logical variables? Often, we deal with measures containing random errors.

A standard scale-position model is written as:

$$X = \mu_{\text{position}} + \sigma_{\text{scale}}\epsilon$$

Where:

- μ is the actual quantity we are interested in estimating
- $\sigma > 0$ represents the noise level. Consequently, $1/\sigma$ represents the precision of our measurement
- ϵ is a random error, treated as a random variable

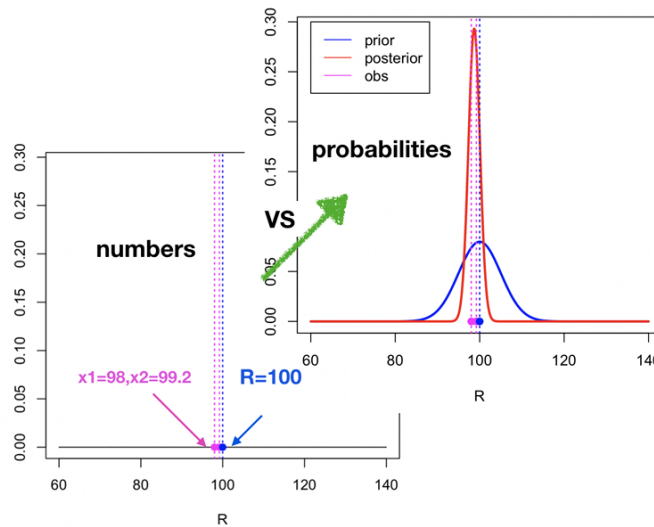
Example: Measuring Resistance

Suppose you are measuring the resistance R of a resistor in a circuit. Your measurements are noisy due to tool imperfections, environmental factors, and manufacturing tolerances.

- **The Likelihood:** The multimeter introduces measurement error. The measurements are approximately normally distributed with a noise level (standard deviation) of 2Ω .
- **The Prior:** The manufacturer specifies the resistance R as 100Ω , but with a tolerance standard deviation of 5Ω .
- **The Data:** You take two measurements yielding $x_1 = 98\Omega$ and $x_2 = 99.2\Omega$.

We can express this formally where $X_i = R + 2\epsilon_i$ and $\epsilon_i \sim \mathcal{N}(0, 1)$. The Bayesian framework seamlessly transitions from discrete 0-1 events to continuous probability distributions, combining our prior knowledge of the resistor with the noisy observed data to yield a continuous posterior distribution.

As we expand our models, we shift from observing isolated events (E) related to discrete hypotheses (H), to observing continuous data vectors ($X \in \mathbb{R}^D$) to infer general parameter spaces ($\vartheta \in \mathbb{R}^M$).



By transitioning from basic numbers to structured probabilities, we build the robust **Bayesian statistical models** that will be explored in depth throughout the rest of this course.

1.4 Probability Review: Random Variables and Densities

To properly engage with Bayesian statistics, we must first establish a rigorous foundation in probability theory. At the heart of modeling random phenomena is the concept of a random variable. A random variable, denoted typically as X , is a numerical outcome of an experiment or random phe-

nomenon whose exact value is not deterministic or completely known *a priori*. Because we cannot precisely predict an individual outcome, a random variable is fully characterized by its probability law (or distribution), which is expressed mathematically by a density function $f_X(x)$. In statistical practice, an *observation* is simply a specific realization (or single outcome) of this random variable, denoted as $X = x$. Typically, we imagine sampling multiple observations from the same underlying population law, resulting in a dataset of realizations $x_1, \dots, x_n$ (conceptually similar to tossing n independent coins).

Important Note

The Nature of Randomness: As humorously noted in classic comic strips featuring a "random number generator", the fundamental problem with randomness is that you can never be absolutely sure an output is truly random just by looking at a sequence of numbers. Instead, we must rely on rigorous probabilistic frameworks to model this inherent uncertainty.

Example: Evaluating Events: CO_2 Emissions

Typically, we are interested in evaluating the probability of an event of the form $A = \{X \in E\}$, where E is a subset of the real numbers $\mathbb{R}$.

For instance, if X represents the per capita CO_2 emission (in tons) for a country in the year 2026, its exact value is currently unknown. Thus, it is entirely sensible to ask what the probability is that emissions will exceed a certain threshold. We formulate this mathematically as $P(A) = P(X > 5.5)$.

1.4.1 Discrete vs. Continuous Distributions

We primarily classify random variables into two distinct types based on the nature of their sample space.

Definition: Types of Random Variables

Discrete Random Variables: A random variable X is discrete if it takes values in a countable set $D = \{x_1, x_2, \dots\}$, meaning $P(X \in D) = 1$. For any set $E \subset \mathbb{R}$, the probability is evaluated using a probability mass function (discrete density), $f(x) = P(X = x)$, such that:

$$P(X \in E) = \sum_{x \in E} f(x)$$

Continuous Random Variables: The random variable X is continuous if, for every subset $E \subset \mathbb{R}$, its probability is determined by integrating a continuous probability density function (PDF) $f(x) \geq 0$:

$$P(X \in E) = \int_E f(x) dx$$

Naturally, the total probability over the entire support must integrate to 1: $\int_{-\infty}^{+\infty} f(x) dx = 1$.

1.4.2 Fundamental Statistical Models

A statistical model is formally described by a family of density functions that depend on a specific parameter, denoted as $f(x|\theta)$. Here, $x \in \mathcal{X} \subset \mathbb{R}^n$ represents the observation (or vector of observations), and $\theta \in \Theta \subset \mathbb{R}^m$ represents the parameter governing the distribution.

Example: Three Essential Statistical Models

Below are three ubiquitous models used in statistics and Bayesian analysis:

1. **Bernoulli Distribution** $Ber(\theta)$: Used for single binary events. The parameter $\theta \in (0, 1)$ represents the probability of achieving a "success".

$$f(x|\theta) = \theta^x(1 - \theta)^{1-x} \quad \text{for } x \in \{0, 1\}$$

2. **Binomial Distribution** $Bin(n, \theta)$: Represents the total number of successes occurring in a sequence of n independent Bernoulli trials.

$$f(x|\theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x} \quad \text{for } x \in \{0, 1, \dots, n\}$$

3. **Normal Distribution** $\mathcal{N}(\mu, \sigma^2)$: A foundational continuous distribution parameterized by a mean μ and variance σ^2 (meaning $\theta = (\mu, \sigma^2)$).

$$f(x|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{for } x \in \mathbb{R}$$

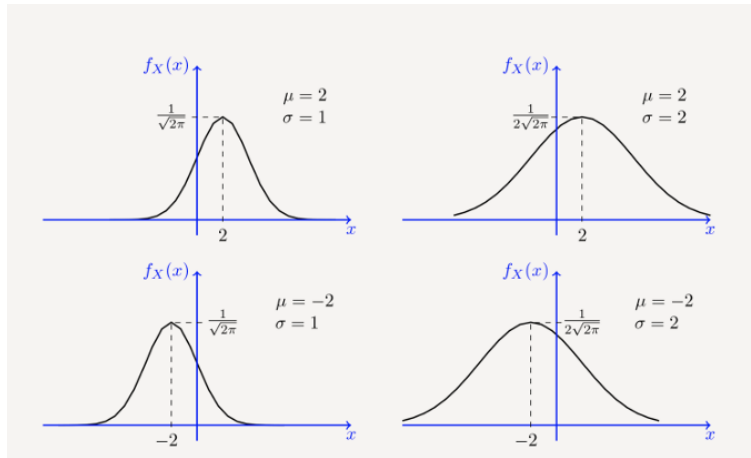


Figure 1: Normal Distribution

1.5 Multivariate Probability and Densities

To expand our Bayesian framework from single variables to more complex inferences, we must utilize joint densities, marginal distributions, and conditional probabilities applied to vectors.

Definition: Joint Distribution

A random vector (X, Y) is completely characterized by its joint distribution. For any region $A \subset \mathbb{R}^2$, the probability that the vector falls within A is:

$$P((X, Y) \in A) = \begin{cases} \sum_{(x,y) \in A} f(x, y) & \text{discrete case} \\ \int \int_{A \subset \mathbb{R}^2} f(x, y) dx dy & \text{continuous case} \end{cases}$$

The joint density $f(x, y)$ must be non-negative and sum or integrate to exactly 1 over the entire space.

Important Note

A Crucial Note on Notation: In standard probability courses, bivariate vectors are typically denoted as (X, Y) or (X_1, X_2) . However, in Bayesian statistics, we are usually dealing with a vector consisting of observed data alongside an unknown parameter. Therefore, moving forward, we will explicitly use the notation (X, ϑ) to represent the joint random vector of observations and parameters. The specific realizations of this vector will be denoted by lowercase letters (x, θ) .

Theorem & Rules: Marginal and Conditional Distributions

If $f(x, \theta)$ is the joint density for the vector (X, ϑ) , we can extract the isolated and conditioned components as follows:

1. Marginalization: We obtain the marginal density of X by integrating (or summing) out the parameter space θ :

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, \theta) d\theta$$

(Replace the integral with a summation over θ for discrete parameter spaces).

2. Conditional Distribution: Recalling the classical probability rule for events where $P(A|B) = P(A \cap B)/P(B)$, the conditional density of the parameter ϑ given an observation $X = x$ is defined as:

$$f_{\vartheta|X}(\theta|x) = \frac{f(x, \theta)}{f_X(x)}$$

Conversely, the conditional density of the observation X given a fixed parameter $\vartheta = \theta$ is:

$$f_{X|\vartheta}(x|\theta) = \frac{f(x, \theta)}{f_{\vartheta}(\theta)}$$

1.6 Bayes' Theorem for Densities

By uniting the definition of conditional probability with the concept of marginalization, we arrive at the continuous counterpart to Bayes' theorem, which serves as the central pillar of Bayesian modeling.

Theorem & Rules: Bayes' Theorem for Densities

Given that the joint density can be expressed as the product of the conditional and marginal distributions, $f(x, \theta) = f_{X|\vartheta}(x|\theta)f_{\vartheta}(\theta)$, Bayes' theorem allows us to invert the conditioning seamlessly:

$$f_{\vartheta|X}(\theta|x) = \frac{f_{X|\vartheta}(x|\theta)f_{\vartheta}(\theta)}{f_X(x)}$$

This formula illustrates the core learning mechanic: it takes our prior distribution for a parameter ($f_{\vartheta}(\theta)$), weights it by the likelihood of the observed data ($f_{X|\vartheta}(x|\theta)$), and normalizes it to produce the updated posterior distribution ($f_{\vartheta|X}(\theta|x)$).

1.7 The Bayesian Core

In Bayesian statistics, a statistical model comprises two fundamental probability density functions:

- The **likelihood**, denoted as $f(x|\theta) = f_{X|\vartheta}(x|\theta)$, which represents the conditional density of the observation X given the parameter θ .
- The **prior**, denoted as $\pi(\theta) = f_{\vartheta}(\theta)$, representing the marginal density and capturing our knowledge or belief about the parameter space before observing the data.

Theorem & Rules: Prior and Posterior Distributions

Given the likelihood and the prior, we can derive the following key distributions:

- **Joint distribution** of the parameter and observation: $f(\theta, x) := f(x|\theta)\pi(\theta)$
- **Marginal distribution** of the observation X : $m(x) = f_X(x) = \int f(x|\theta)\pi(\theta)d\theta$
- **Posterior distribution** of θ : Derived via Bayes' theorem, it is expressed as $\pi(\theta|x) = \frac{f_{\vartheta|X}(\theta|x)}{\int_{\Theta} f_{\vartheta|X}(u|x)\pi(u)du}$

The posterior distribution forms the absolute center of Bayesian inference. It is mathematically proportional to the likelihood multiplied by the prior ($\pi(\theta|x) \propto f(x|\theta)\pi(\theta)$). This distribution operates conditionally upon the observations, allowing us to simultaneously integrate prior information and the new knowledge brought by the data x . This unified framework provides a complete scope for inference, which can be applied to predictions, parameter estimation, deriving credible regions, and hypothesis testing.

1.7.1 Frequentist vs. Bayesian Perspectives

A central philosophical distinction separates Frequentist and Bayesian approaches:

- **Frequentist View:** The parameter θ is considered an unknown but fixed value. Probability exists exclusively in the data and the sampling process.
- **Bayesian View:** The parameter θ is treated as a random variable. Probability is utilized to model the learning process itself, which is inherently probabilistic.

Important Note

The Bayesian perspective requires a semantic shift from thinking of θ as "unknown" to treating it as "random". Inference is directly based on the distribution of θ conditional on x , effectively actualizing prior knowledge by extracting the information contained in the observation.

2 Lesson 2: Models and Conjugate Priors

2.1 Independent and Identically Distributed (i.i.d.) Observations

Before diving into conjugate models, let us recap the general Bayesian model structure, which requires a likelihood for the observations $x_{1:n} = (x_1, \dots, x_n)$ and a prior $\pi(\theta)$. The core goal is to compute the posterior: $\pi(\theta|x) \propto f(x|\theta)\pi(\theta)$.

Important Note

Notation Warning: In probability theory, it is common to distinguish a random variable (X) from its realization (x). In applied statistics, we often use the same lowercase letter (x) to denote both.

Definition: i.i.d. Observations

A vector $(x_1, \dots, x_n)$ is an i.i.d. sample from $f(x|\theta)$ if the joint likelihood factorizes into the product of the marginal densities. Mathematically, $f(x_1, \dots, x_n|\theta) = f(x_1|\theta) \cdots f(x_n|\theta)$.

For an i.i.d. sample, the posterior distribution expands to:

$$\pi(\theta|x_1, \dots, x_n) = \frac{f(x_n|\theta) \cdots f(x_1|\theta)\pi(\theta)}{\int f(x_n|u) \cdots f(x_1|u)\pi(u)du}$$

2.1.1 The Normal-Normal Model

Theorem & Rules: Normal-Normal Conjugacy

Assume observations $x_1, \dots, x_n$ are conditionally i.i.d. following a Normal distribution $\mathcal{N}(\mu, \sigma^2)$, where the variance σ^2 is assumed known. If we place a Normal prior on the mean $\mu \sim \mathcal{N}(\mu_0, \tau^2)$, the posterior distribution of μ is also Normal: $\mathcal{N}(\mu_n, \sigma_n^2)$.

The updated posterior parameters are defined as:

$$\mu_n = \frac{\sigma^2}{\sigma^2 + n\tau^2}\mu_0 + \frac{n\tau^2}{\sigma^2 + n\tau^2} \left(\frac{1}{n} \sum_{i=1}^n x_i \right)$$
$$\sigma_n^2 = \frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2}$$

Example: Resistor Measurement Diagnosis

Recall the scenario where we measure a resistor:

- **Prior:** Manufacturer specs indicate the resistance is $\mathcal{N}(100, 5^2)$, giving $\mu_0 = 100\Omega$ and $\tau = 5\Omega$.
- **Likelihood (Noise):** The multimeter has a measurement noise standard deviation of $\sigma = 2\Omega$.
- **Observations:** We take $n = 2$ measurements: $x_1 = 98\Omega$ and $x_2 = 99.2\Omega$.

By applying the Normal-Normal posterior equations, we can compute the exact updated normal distribution, blending our initial expectations with the empirical data.

2.2 Predictive Distributions

Bayesian learning also enables us to forecast future observations based on what we have already seen.

Definition: Predictive Distribution

The predictive distribution of a new observation x_{new} given observed data x_{obs} is the conditional distribution of x_{new} given x_{obs} . It depends on the joint distribution of (x_{new}, x_{obs}) , typically established by the conditional predictive distribution $g(x_{new}|\theta, x_{obs})$.

To compute the predictive distribution unconditionally of θ , we marginalize over the posterior:

$$g(x_{new}|x) = \int g(x_{new}|\theta, x)\pi(\theta|x)d\theta$$

Theorem & Rules: Predictive Distribution for i.i.d. Data

If $x_1, \dots, x_n, x_{n+1}$ are conditionally i.i.d. given θ with density $f(x_i|\theta)$, the predictive distribution simplifies because $g(x_{new}|\theta, x) = f(x_{new}|\theta)$. The formula becomes:

$$g(x_{n+1}|x_1, \dots, x_n) = \int f(x_{new}|\theta)\pi(\theta|x)d\theta$$

2.2.1 Normal-Normal Model: Predictive Distribution

Building upon the Normal-Normal model where observations $x_1, \dots, x_n$ are i.i.d. following $\mathcal{N}(\mu, \sigma^2)$ and the prior is $\mu \sim \mathcal{N}(\mu_0, \tau^2)$, we know the posterior distribution is $\mu|x_{1:n} \sim \mathcal{N}(\mu_n, \sigma_n^2)$.

Using this updated knowledge, the predictive distribution for a new, unseen observation x_{n+1} also follows a normal distribution:

$$x_{n+1} \sim \mathcal{N}(\mu_n, \sigma_n^2 + \sigma^2)$$

Example: FBI Larcenies Dataset

Let $X_1, \dots, X_n$ represent the relative changes in reported larcenies between 1991 and 1995 for the 90 most populous US counties (Source: FBI). We might assume that these relative changes roughly follow a normal distribution.

2.3 Review of Probability: Expectations and Variances

To effectively summarize posterior distributions and evaluate Bayesian models, we must review how to compute the expected value and variance of random variables.

Definition: Expectation / Mean Value

The expectation of a random variable X , denoted as $\mathbb{E}[X]$, is defined as:

$$\mathbb{E}[X] := \begin{cases} \sum_x x f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} x f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

For any real-valued function $g(X)$, its expected value $\mathbb{E}[g(X)]$ is computed by substituting x with $g(x)$ inside the summation or integral.

Definition: Variance

Let $m = \mathbb{E}[X]$. The variance of X , measuring the dispersion of the variable around its mean, is defined as $Var(X) := \mathbb{E}[(X - m)^2]$.

$$Var(X) = \begin{cases} \sum_x (x - m)^2 f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} (x - m)^2 f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Theorem & Rules: Moments of Common Statistical Models

- **Normal:** If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $\mathbb{E}[X] = \mu$ and $Var(X) = \sigma^2$
- **Bernoulli:** If $X \sim \mathcal{Ber}(\theta)$, then $\mathbb{E}[X] = \theta$ and $Var(X) = \theta(1 - \theta)$
- **Binomial:** If $X \sim \mathcal{Bin}(n, \theta)$, then $\mathbb{E}[X] = n\theta$ and $Var(X) = n\theta(1 - \theta)$

2.4 The Posterior Mean

Once the posterior distribution of a real-valued parameter θ is derived, we often want to express our updated inference using a single, representative number.

Theorem & Rules: Posterior Mean

The posterior mean is defined as the conditional expectation of θ given the observations x :

$$\mathbb{E}[\theta|x] = \begin{cases} \sum_{\theta} \theta \pi(\theta|x) & \text{in the discrete case} \\ \int \theta \pi(\theta|x) d\theta & \text{in the continuous case} \end{cases}$$

Example: Posterior Mean of the Normal Model

Consider again our model where $x_1, \dots, x_n$ are conditionally i.i.d. $\mathcal{N}(\mu, \sigma^2)$ and the prior is $\mu \sim \mathcal{N}(\mu_0, \tau^2)$. The posterior distribution of μ given the data $x_{1:n}$ is $\mathcal{N}(\mu_n, \sigma_n^2)$. Because the mean of a normal distribution is its primary location parameter, the posterior mean simplifies directly to:

$$\mathbb{E}[\mu|x_1, \dots, x_n] = \mu_n$$

2.5 Conjugate Models

The prior distribution is the definitive key to Bayesian inference. However, in practical applications, the available prior information is seldom precise enough to lead to an exact, indisputable determination of the prior distribution. As a guiding rule in Bayesian statistics: there is no such thing as *the* singular correct prior distribution!. To build robust and mathematically manageable models, we frequently rely on *conjugate priors*.

Definition: Conjugate Prior

A parameterized family $\mathcal{F}$ of probability distributions on a parameter space Θ is called **conjugate** for a likelihood function $f(x|\theta)$ if, for every prior $\pi \in \mathcal{F}$, the resulting posterior distribution $\pi(\theta|x)$ also belongs to $\mathcal{F}$.

When using conjugate priors, switching from the prior to the posterior distribution is elegantly reduced to simply updating the corresponding parameters.

Important Note

Why do we heavily utilize conjugate models?

- **Tractability and Simplicity:** They provide clean, closed-form analytical solutions that are computationally efficient.
- **Preservation of Structure:** The mathematical structure of $\pi(\theta)$ remains consistent before and after observing data.
- **Virtual Past Observations:** The hyperparameters of a conjugate prior can often be interpreted intuitively as introducing a specific number of "virtual" past observations.
- **Linearity of Estimators:** They frequently lead to highly interpretable, linear estimators for parameters.

Moving forward, we will detail several foundational conjugate models essential to the Bayesian core:

1. The Beta-Binomial Model
2. The Gamma-Poisson Model
3. The Normal-Normal Model (which we have previewed)
4. The Normal-Normal-Inverse Gamma Model

2.5.1 The Beta-Binomial Model

To understand the Beta-Binomial model, we first need to define its foundational distributions.

Definition: Beta Distribution

A generalization of the continuous uniform distribution on $[0, 1]$ is the Beta distribution, denoted as $\mathcal{Be}(\alpha, \beta)$. Its density function is defined for $\theta \in (0, 1)$ with parameters $\alpha > 0$ and $\beta > 0$ as:

$$\pi(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the Beta function.

Definition: Binomial Distribution

Let X be the number of successes that will occur in a sequence of n independent Bernoulli trials, each with success probability θ . The probability mass function of $X \sim \text{Bin}(n, \theta)$ is:

$$f(x|\theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x} \quad \text{for } x = 0, 1, \dots, n$$

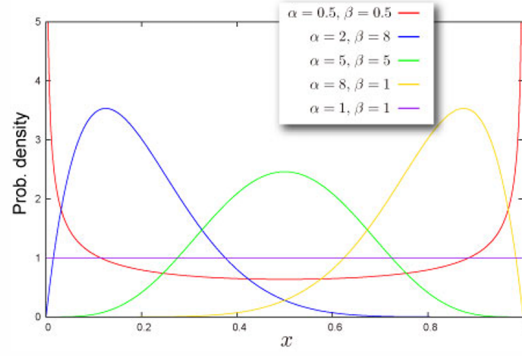


Figure 2: Beta Distribution for different values of α and β

Theorem & Rules: Beta-Binomial Conjugacy

In the Beta-Binomial model, we observe data $x \sim \text{Bin}(n, \theta)$ and assign a Beta prior $\theta \sim \text{Be}(\alpha, \beta)$. The joint distribution of the data and parameter is $f(x, \theta) = \binom{n}{x} \frac{1}{B(\alpha, \beta)} \theta^{\alpha+x-1} (1-\theta)^{\beta+n-x-1}$. By applying Bayes' theorem (dividing the joint distribution by the marginal distribution $m(x)$), the posterior distribution of θ is found to be proportional to:

$$\pi(\theta|x) \propto \theta^{\alpha+x-1} (1-\theta)^{\beta+n-x-1}$$

This means the posterior is exactly a Beta distribution with updated parameters: $\theta|x \sim \text{Be}(\alpha + x, \beta + n - x)$.

Important Note

A Bayes estimator $\hat{\theta}$ for the parameter θ is typically the mean of the posterior distribution. For the Beta-Binomial model, the posterior mean evaluates to:

$$\hat{\theta} = \mathbb{E}[\theta|x] = \frac{\alpha + x}{\alpha + \beta + n}$$

Notice how this estimate mathematically bridges the prior mean ($\frac{\alpha}{\alpha+\beta}$) and the empirical evidence provided by the new data.

2.5.2 The Gamma-Poisson Model

Next, we explore models suitable for count data or events occurring over a specific interval, utilizing the Poisson and Gamma distributions.

Definition: Poisson Distribution

Let X represent the number of events (e.g., customers calling a help center in a day). X is a Poisson random variable, denoted $\mathcal{Poi}(\lambda)$ (or sometimes $\mathcal{Poi}(\theta)$), if its discrete probability mass function is:

$$f(x|\lambda) = \frac{\lambda^x}{x!} e^{-\lambda} \quad \text{for } x = 0, 1, 2, \dots$$

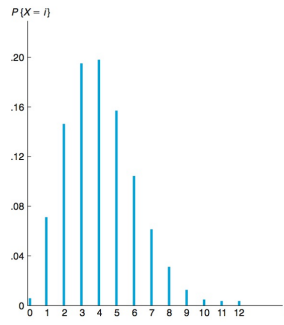


Figure 3: Poisson Distribution for different values of λ

Definition: Gamma Distribution

A continuous random variable λ follows a Gamma distribution $\mathcal{Ga}(\alpha, \beta)$ if its probability density function is:

$$\pi(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \mathbb{I}_{\{\lambda>0\}}$$

Here, $\alpha > 0$ serves as the shape parameter and $\beta > 0$ as the rate parameter. Note that $1/\beta$ acts as the scale parameter.

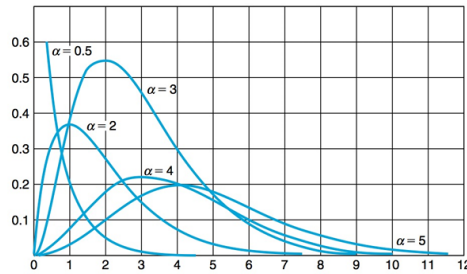


Figure 4: Gamma Distribution for different values of α and $\beta = 1$

Theorem & Rules: Gamma-Poisson Conjugacy

Assume we have conditionally independent observations $x_i \sim \mathcal{Poi}(\lambda)$ and we place a Gamma prior $\lambda \sim \mathcal{Ga}(\alpha, \beta)$. The joint likelihood of the sample is proportional to $e^{-n\lambda} \lambda^{\sum x_i}$. Multiplying this likelihood by the prior yields the posterior distribution of λ :

$$\pi(\lambda|x_1, \dots, x_n) \propto e^{-(\beta+n)\lambda} \lambda^{\alpha + \sum_{i=1}^n x_i - 1}$$

This functional form corresponds to a new Gamma distribution $\mathcal{Ga}(\alpha_n, \beta_n)$ with the updated parameters:

$$\alpha_n = \alpha + \sum_{i=1}^n x_i, \quad \beta_n = \beta + n$$

Important Note

Estimation and Prediction in the Gamma-Poisson Model:

- **Posterior Mean (Bayes Estimator):** The expected value of the posterior is $\hat{\lambda}_n = \frac{\alpha_n}{\beta_n} = \frac{\alpha + \sum x_i}{\beta + n}$. This estimator naturally breaks down into a weighted sum of the prior mean (α/β) and the Maximum Likelihood (ML) estimator $(\bar{x} = \frac{1}{n} \sum x_i)$.
- **Predictive Distribution:** The predictive distribution for a completely new, unseen observation x_{n+1} , conditional on the past data $x_{1:n}$, takes the form of a Negative Binomial distribution: $x_{n+1}|x_{1:n} \sim \mathcal{NB}(\alpha_n, \beta_n(1 + \beta_n)^{-1})$.
- **Predictive Moments:** The predictive mean is $\mathbb{E}[x_{n+1}|x_{1:n}] = \frac{\alpha_n}{\beta_n}$. The predictive variance is slightly larger due to the uncertainty in the parameter: $\text{Var}(x_{n+1}|x_{1:n}) = \frac{\alpha_n}{\beta_n} (1 + \frac{1}{\beta_n})$.

2.5.3 The Normal-Normal Model (Recap)

Before introducing complexity, it is helpful to briefly recap the conjugate Normal-Normal model where the variance σ^2 is assumed to be known.

Theorem & Rules: Normal-Normal Conjugacy Update

If our likelihood is given by conditionally independent observations $x_i \sim \mathcal{N}(\mu, \sigma^2)$ and we set a normal prior on the mean $\mu \sim \mathcal{N}(\mu_0, \tau^2)$, the joint likelihood and prior combine to form a new normal posterior distribution.

Specifically, the posterior is $\mu|x_1, \dots, x_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$, where the updated parameters are:

$$\mu_n = \frac{\sigma^2}{\sigma^2 + n\tau^2}\mu_0 + \frac{n\tau^2}{\sigma^2 + n\tau^2}\frac{1}{n}\sum_{i=1}^n x_i$$
$$\sigma_n^2 = \frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2}$$

The posterior mean, serving as our Bayes estimator, is exactly $\mathbb{E}[\mu|x_1, \dots, x_n] = \mu_n$. Furthermore, the predictive distribution for a new observation x_{n+1} is given by $x_{n+1} \sim \mathcal{N}(\mu_n, \sigma_n^2 + \sigma^2)$.

2.5.4 The Full Normal Model: Unknown Mean and Variance

In practical applications, it is incredibly rare to know the exact variance σ^2 of a population. Therefore, we must extend our model to the **Full Normal Model**, where both the mean μ and the variance σ^2 are treated as unknown parameters. To maintain conjugacy for the parameter vector $\theta = (\mu, \sigma^2)$, we introduce the Normal-Inverse Gamma prior.

Definition: Normal-Inverse Gamma Prior

The Normal-Inverse Gamma model (often framed as Normal-Gamma when parameterized with precision σ^{-2}) relies on a joint prior distribution. We specify it hierarchically:

1. **Prior for the variance:** $\sigma^2 \sim \mathcal{IG}(\alpha_0, \beta_0)$ (or equivalently, precision $\sigma^{-2} \sim \mathcal{G}(\alpha_0, \beta_0)$).
2. **Conditional prior for the mean:** $\mu|\sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2/k_0)$.

Here, α_0, β_0, μ_0 , and k_0 are hyperparameters. The parameter k_0 can be interpreted as the number of "virtual" prior observations determining the precision of our prior mean μ_0 .

Theorem & Rules: Posterior of the Full Normal Model

Given the likelihood $f(x_1, \dots, x_n | \mu, \sigma^2) \propto (\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} [n(\mu - \bar{x}_n)^2 + s_x^2] \right\}$ (where $s_x^2 = \sum_{i=1}^n (x_i - \bar{x}_n)^2$ is the sample sum of squares), multiplying by the Normal-Inverse Gamma prior yields a posterior distribution belonging to the exact same family.

The posterior distribution of $(\mu, \sigma^2) | x_1, \dots, x_n$ is Normal-Inverse Gamma with the following updated parameters:

$$\begin{aligned}\alpha_n &= \alpha_0 + \frac{n}{2} \\ k_n &= k_0 + n \\ \mu_n &= \frac{k_0 \mu_0 + n \bar{x}_n}{k_0 + n} \\ \beta_n &= \beta_0 + \frac{s_x^2}{2} + \frac{n k_0 (\bar{x}_n - \mu_0)^2}{2(n + k_0)}\end{aligned}$$

2.5.5 Marginal and Predictive Distributions

While the joint posterior provides a complete picture, we are often interested strictly in the mean μ , regardless of the variance, or in predicting future observations. Integrating out the variance σ^2 leads us to the Student's t-distribution.

Important Note

The Student's t-Distribution: Let $T \sim \mathcal{T}_r$ be a standard Student's t-distribution with r degrees of freedom. We can define a generalized version $T^* = sT + m$, which follows $\mathcal{T}_r(m, s^2)$, where m is a position (location) parameter and s is a scale parameter. Due to the symmetry of the distribution, if $t_{\alpha/2, r}$ is the $\alpha/2$ quantile of T , then the probability bounded by these quantiles is exactly $\mathbb{P}(-t_{\alpha/2, r} < T < t_{\alpha/2, r}) = 1 - \alpha$.

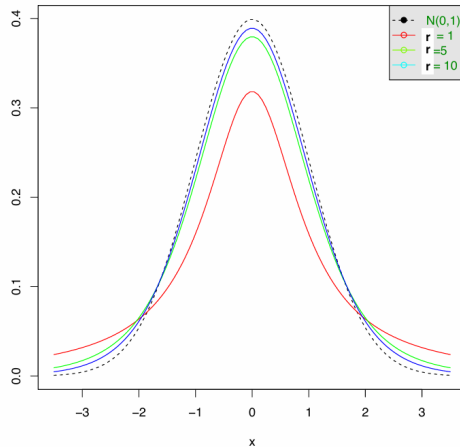


Figure 5: Student Distribution for different values of r

Theorem & Rules: Marginal Posterior and Predictive Distributions

If the joint prior (or posterior) is Normal-Inverse Gamma, the marginal distribution of the mean μ (integrating out σ^2) follows a generalized Student's t-distribution:

$$\mu \sim \mathcal{T}_{2\alpha_0} \left(\mu_0, \frac{\beta_0}{\alpha_0 k_0} \right)$$

.

By applying this to our updated posterior parameters, the **marginal posterior of μ** is $\mathcal{T}_{2\alpha_n} \left(\mu_n, \frac{\beta_n}{\alpha_n k_n} \right)$

Furthermore, the **predictive distribution** for a new observation x_{n+1} , given the observed data $x_{1:n}$, is also a Student's t-distribution:

$$x_{n+1} | x_1, \dots, x_n \sim \mathcal{T}_{2\alpha_n} \left(\mu_n, \frac{\beta_n(k_n + 1)}{\alpha_n k_n} \right)$$

2.6 Bayesian Decision Theory and Estimation

The primary purpose of most inferential studies is to provide the statistician or client with a concrete decision $d \in D$, where D is the decision space. When our goal is parameter estimation, the decision is simply to "select a parameter as your estimate," meaning $D = \Theta$. To evaluate how good a decision is, we require an evaluation criterion known as a loss function.

Definition: Loss Function and Bayes Estimator

A **loss function** is a mapping $L(\theta, d) : \Theta \times D \rightarrow \mathbb{R}_+$. It quantifies the penalty or loss incurred when the true parameter is θ and the chosen decision or estimate is d .

The statistical decision is a function of the observed sample, denoted $\delta(x) : \mathcal{X} \rightarrow D$. A **Bayes estimator** associated with a prior distribution and a specific loss function is the decision rule $\delta(x)$ that minimizes the *expected posterior loss*:

$$\delta(x) = \arg \min_{\delta} \mathbb{E}[L(\theta, \delta(x)) | x]$$

Theorem & Rules: Common Loss Functions and their Estimators

Assuming $\Theta \subset \mathbb{R}$, different choices of loss functions naturally lead to different optimal Bayes estimators:

- **Quadratic Loss** $L(\theta, d) = (\theta - d)^2$: Minimizing the expected quadratic loss yields the **posterior mean**:

$$\theta_{mean}(x) = \mathbb{E}[\theta|x] = \frac{\int_{\Theta} \theta f(x|\theta)\pi(\theta)d\theta}{\int_{\Theta} f(x|\theta)\pi(\theta)d\theta}$$

- **Absolute Value Loss** $L(\theta, d) = |\theta - d|$: Minimizing the expected absolute loss yields the **posterior median**:

$$\theta_{med}(x) = \text{median of } \pi(\theta|x)$$

- **Zero-One Loss** (often associated with "no loss" mathematically): This yields the **Maximum A Posteriori (MAP)** estimator, which is the mode of the posterior distribution:

$$\theta_{MAP}(x) = \arg \max_{\theta} f(x|\theta)\pi(\theta)$$

2.6.1 Risk Functions (Optional Note)

While Bayesian inference relies on minimizing the posterior expected loss $\rho(\pi, \delta) = \mathbb{E}[L(\theta, \delta(x))|x]$, frequentists often use the *risk function* $R(\theta, \delta) = \int_{\mathcal{X}} L(\theta, \delta(x))f(x|\theta)dx$. Because it is generally impossible to uniformly minimize the risk function across all possible values of θ , the risk function alone cannot define an optimal estimator without further constraints.

2.7 Interval Estimation: Credible Regions

Beyond point estimation, we often want to provide a range of plausible values for our parameter. In Bayesian statistics, this is done using *credible regions*.

Definition: Bayesian Credible Region

A subset $C_{\alpha}(x) \subset \Theta$ is a Bayesian credible region of level α for the parameter ϑ if the posterior probability that ϑ falls within this region is exactly $1 - \alpha$:

$$Prob\{\vartheta \in C_{\alpha}(X)|X = x\} = \pi(C_{\alpha}(x)|x) = 1 - \alpha$$

Typically, standard values for α are 0.05 or 0.01, corresponding to 95% or 99% credible regions.

Important Note

Confidence vs. Credibility

It is crucial to understand that a Bayesian credible region has a fundamentally different interpretation than a frequentist confidence interval, even if they occasionally yield the exact same numbers.

- **Bayesian (Credible Interval):** "Given our observed data, there is a 95% probability that the true value of the parameter falls within $C_{0.05}$." The parameter is treated as random, and the interval is fixed by the data.
- **Frequentist (Confidence Interval):** "There is a 95% probability that when I compute a confidence interval from data generated by this process, the computed interval will contain the true, fixed mean." The parameter is fixed, and the interval bounds are random variables.

To ensure our credible region is as tight as possible, we often look for the **Highest Posterior Density (HPD) regions**. An HPD region is defined as $C_\alpha = \{\theta : \pi(\theta|x) > k_\alpha\}$, where k_α is chosen to achieve the desired $1 - \alpha$ probability mass. HPD regions are optimal because they provide the smallest possible volume for a given probability level.

2.7.1 Quantile

We say that the number q_α is the $100(1 - \alpha)$ percentile (or α quantile) of a random variable X if

$$P\{X > q_\alpha\} = \alpha$$

If the distribution of X is symmetric (and continuous)

$$P\{-q_{\alpha/2} \leq X \leq q_{\alpha/2}\} = 1 - \alpha.$$

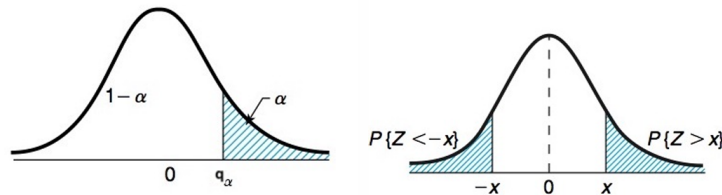


Figure 6: Quantile-based credible interval

2.7.2 Credible Intervals via Quantiles

For continuous, symmetric posterior distributions, we can easily construct a credible interval using quantiles.

If the transformed posterior distribution $\tilde{\theta} := (\theta - c)/b$ is symmetric (for suitable centering constant c and normalizing constant b), and $q_{\alpha/2}$ is the $\alpha/2$ quantile of $\tilde{\theta}$, then:

$$C_\alpha = (c - q_{\alpha/2}b, \quad c + q_{\alpha/2}b)$$

is a valid credible interval of level α . Because of the symmetry, this quantile-based interval is inherently also a Highest Posterior Density (HPD) region.

2.7.3 Intervals for the Full Normal Model

Let us apply this quantile concept to the Full Normal model (Normal-Inverse Gamma prior). We previously established that the marginal posterior of the mean μ is a generalized Student's t-distribution: $\mathcal{T}_{2\alpha_n}(\mu_n, \beta_n/(\alpha_n k_n))$.

Here, μ_n is the centering parameter, and we can define the scale parameter as $\omega_n := \sqrt{\beta_n/(\alpha_n k_n)}$.

Theorem & Rules: Credible Intervals - Full Normal Model

A Bayesian credible interval at level $\alpha \in (0, 1)$ for the mean μ in the full normal model is given by:

$$C_\alpha^\pi = [\mu_n - t_{\alpha/2, 2\alpha_n} \omega_n, \quad \mu_n + t_{\alpha/2, 2\alpha_n} \omega_n]$$

where $t_{\alpha/2, r}$ is the $\alpha/2$ -quantile of a standard Student's t-distribution with $r = 2\alpha_n$ degrees of freedom.

For comparison, the standard *frequentist* two-sided confidence interval for the parameter μ (when variance is unknown) is:

$$\left[\bar{x}_n - t_{\alpha/2, n-1} \frac{s_x}{\sqrt{n(n-1)}}, \quad \bar{x}_n + t_{\alpha/2, n-1} \frac{s_x}{\sqrt{n(n-1)}} \right]$$

where $s_x = \sqrt{\sum_{i=1}^n (x_i - \bar{x}_n)^2}$.

Example: Normaldata Confidence Region

Using the `normaldata` dataset (relative changes in reported larcenies), we can compute a 95% confidence region for μ .

Assume prior hyperparameters $\mu_0 = 0, k_0 = 0.1, a_0 = 0.1, b_0 = 0.1$. By plugging these into our posterior update formulas and utilizing the t -quantiles, we derive the interval boundaries. The critical analytical question then becomes: based on this interval, is it acceptable to report a significant decrease in the number of larcenies between 1991 and 1995? (This naturally leads us into hypothesis testing).

2.8 Hypothesis Testing

Deciding about the validity of assumptions or restrictions on a parameter θ from observed data is a classic problem in statistical inference. We typically formulate this by partitioning the parameter

space into two mutually exclusive sets, representing a null hypothesis (H_0) and an alternative hypothesis (H_1):

$$H_0 : \theta \in \Theta_0 \quad \text{versus} \quad H_1 : \theta \notin \Theta_0$$

The outcome of this decision process is binary: we either accept the hypothesis (coded by 1) or reject it (coded by 0), meaning our decision space is $D = \{0, 1\}$.

2.8.1 The 0-1 Loss Function

Theorem & Rules: The 0-1 Loss Function and Bayes Estimator

A rudimentary yet foundational loss function for hypothesis testing is the 0 – 1 loss, defined as:

$$L(\theta, d) = \begin{cases} 1 - d & \text{if } \theta \in \Theta_0 \\ d & \text{otherwise} \end{cases}$$

By minimizing the expected posterior loss, we derive the associated Bayes estimator (decision rule). We accept the null hypothesis if its posterior probability exceeds 50%:

$$\delta_\pi(x) = \begin{cases} 1 & \text{if } \pi(\theta \in \Theta_0|x) > \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

2.8.2 Testing in the Full Normal Model

Consider the full normal model where we wish to test if the mean μ is less than or equal to a specific threshold m_0 (i.e., $\Theta_0 = \{\mu \leq m_0\}$).

Recall that the marginal posterior of μ is a generalized Student's t-distribution: $\mathcal{T}_{2\alpha_n}(\mu_n, \omega_n^2)$, where $\omega_n = \sqrt{\beta_n/(\alpha_n k_n)}$. We can compute the posterior probability of the null hypothesis by standardizing the variable:

$$\pi\{\Theta_0|x_{1:n}\} = P\{\mu \leq m_0|x_{1:n}\} = P\left\{\frac{\mu - \mu_n}{\omega_n} \leq \frac{m_0 - \mu_n}{\omega_n} \middle| x_{1:n}\right\}$$

Since the standardized term follows a standard Student's t-distribution with $2\alpha_n$ degrees of freedom, the probability is simply evaluated using its cumulative distribution function (CDF), denoted $F_{t,2\alpha_n}$:

$$\pi\{\Theta_0|x_{1:n}\} = F_{t,2\alpha_n}\left(\frac{m_0 - \mu_n}{\omega_n}\right)$$

2.9 The Bayes Factor

When dealing with hypothesis testing, the posterior probability $\pi\{\Theta_i|x\}$ is often highly sensitive to the prior weights assigned to each hypothesis ($w_0 = \pi(\Theta_0)$ and $w_1 = \pi(\Theta_1)$). To measure the evidence provided *exclusively by the data*, independent of these initial prior weights, we use the Bayes Factor.

Definition: Bayes Factor

The Bayes factor B_{10}^π is the ratio of the posterior odds to the prior odds of the hypotheses:

$$B_{10}^\pi = \frac{\pi(\theta \in \Theta_1|x)/\pi(\theta \in \Theta_0|x)}{\pi(\theta \in \Theta_1)/\pi(\theta \in \Theta_0)} = \frac{m_1(x)}{m_0(x)}$$

where $m_1(x)$ and $m_0(x)$ are the marginal densities of the data under hypothesis 1 and 0, respectively.

Important Note

[Jeffreys' Scale of Evidence] To interpret the magnitude of the Bayes Factor B_{10}^π , statisticians often refer to Jeffreys' scale using the base-10 logarithm:

- $\log_{10}(B_{10}^\pi) < 0$: No evidence against H_0 .
- $0 \leq \log_{10}(B_{10}^\pi) \leq 0.5$: Evidence against H_0 is poor.
- $0.5 < \log_{10}(B_{10}^\pi) \leq 1$: Evidence against H_0 is substantial.
- $1 < \log_{10}(B_{10}^\pi) \leq 2$: Evidence against H_0 is strong.
- $\log_{10}(B_{10}^\pi) > 2$: Evidence against H_0 is decisive.

2.10 Point Null Hypotheses and Dirac Masses

A significant mathematical difficulty arises when testing a *point null hypothesis*, such as $H_0 : \theta = \theta_0$.

If our prior distribution $\pi(\theta)$ is absolutely continuous, the probability of any single specific point is exactly zero: $\pi(\theta = \theta_0) = 0$. Consequently, the posterior probability $\pi(\theta = \theta_0|x)$ will also invariably be zero, making standard testing impossible.

Theorem & Rules: Dirac Mass Prior

Testing a point null hypothesis requires a structural modification of the prior distribution to ensure that $\pi(\Theta_0) > 0$. We achieve this by mixing a continuous prior $\pi_1(\theta)$ with a point mass (Dirac delta function δ_{θ_0}) at the hypothesized value:

$$\pi(\theta) = \rho\delta_{\theta_0}(\theta) + (1 - \rho)\pi_1(\theta)$$

where ρ represents the prior probability w_0 assigned to the null hypothesis (e.g., $\rho = 0.5$).

Under this mixed prior, the posterior probability of the null hypothesis correctly updates based on the likelihood evaluated at θ_0 and the marginal likelihood $m_1(x)$ under the continuous alternative:

$$\pi(\Theta_0|x) = \frac{f(x|\theta_0)\rho}{f(x|\theta_0)\rho + (1 - \rho)m_1(x)}$$

2.11 Improper Priors

In cases where we lack substantive prior knowledge, we might attempt to use a completely flat or uniform prior to express ignorance. Historically, Laplace's Principle of Insufficient Reason suggested assigning equal probability to all possible states.

When extended to a continuous parameter space like the real line $\mathbb{R}$, a constant prior ($\pi(\theta) \propto 1$) corresponds to the Lebesgue measure.

Definition: Improper Prior

An improper prior is a σ -finite measure π that fails to integrate to a finite number over the parameter space Θ :

$$\int_{\Theta} \pi(\theta) d\theta = +\infty$$

Formally, π cannot be interpreted as a true probability distribution.

Important Note

Validation of Improper Priors: While an improper prior is not a valid probability distribution, the resulting posterior distribution calculated via Bayes' formula can still be perfectly valid. The requirement is that the marginal integral must be finite:

$$\int_{\Theta} f(x|\theta) \pi(\theta) d\theta < +\infty$$

If this condition is met, the posterior $\pi(\theta|x)$ is a proper probability distribution.

2.11.1 The Full Normal Model with an Improper Prior

A standard non-informative, improper prior for the full normal model is $\pi(\mu, \sigma^2) = 1/\sigma^2$. Under this prior, the posterior marginal distribution of μ simplifies cleanly to a Student's t-distribution:

$$\mu|\bar{x}, s_x^2 \sim \mathcal{T}_{n-1}(\bar{x}, s_x^2)$$

where the density is proportional to $[s_x^2 + n(\bar{x} - \mu)^2]^{-n/2}$.

Remarkably, the Bayesian credible interval derived from this specific improper prior exactly matches the classic frequentist confidence interval for the mean:

$$\left[\bar{x} - t_{\alpha/2, n-1} \frac{s_x}{\sqrt{n(n-1)}}, \quad \bar{x} + t_{\alpha/2, n-1} \frac{s_x}{\sqrt{n(n-1)}} \right]$$

2.12 Exponential Families (Optional)

Many of the conjugate models we have discussed share an underlying mathematical architecture known as the Exponential Family.

Definition: Exponential Family

A family of distributions is an exponential family of dimension k if its density (or mass) function can be factored as:

$$f(x|\theta) = C(\theta)h(x) \exp\{R(\theta) \cdot T(x)\}$$

When the parameterization is $\Theta \subset \mathbb{R}^k$ and $X \subset \mathbb{R}^k$, the natural (canonical) form is written as:

$$f(x|\theta) = h(x) \exp\{\theta \cdot x - \Psi(\theta)\}$$

where $\Psi(\theta) = \log(1/C(\theta))$ and $C(\theta)^{-1}$ is the normalizing constant required to ensure the distribution integrates to 1.

Theorem & Rules: Conjugacy in Exponential Families

The primary analytical advantage of exponential families is that they naturally allow for conjugate priors. For a likelihood in canonical form, the conjugate prior takes the structure:

$$\pi(\theta|\mu, \lambda) = K(\mu, \lambda) \exp\{\theta \cdot \mu - \lambda \Psi(\theta)\} \quad (\text{for } \lambda > 0)$$

When a single observation x is drawn, the posterior distribution retains the exact same functional form, with elegantly simple parameter updates:

$$\begin{aligned}\mu &\rightarrow \mu + x \\ \lambda &\rightarrow \lambda + 1\end{aligned}$$

This structural elegance is what underpins the conjugacy rules for the Normal, Poisson, Gamma, Binomial, Negative Binomial, and Dirichlet models used extensively throughout Bayesian learning.

2.12.1 Summary of Standard Exponential Families and Conjugacy

The fundamental analytical advantage of exponential families is the existence of conjugate priors. The following tables concisely summarize the likelihoods, conjugate priors, and the resulting posterior distributions for some of the most widely used statistical models.

Important Note

Interpretation of Updates: Notice how elegantly the parameters transition in these exponential families. For instance, in the Gamma-Poisson model, observing a data point x simply requires adding the observation to the shape parameter ($\alpha \rightarrow \alpha + x$) and incrementing the rate parameter by the number of observations ($\eta \rightarrow \eta + 1$). This showcases the computational simplicity that conjugate models provide.

Model Family	Likelihood $f(x \theta)$	Prior $\pi(\theta)$	Posterior $\pi(\theta x)$
Normal (unknown mean)	$\mathcal{N}(\theta, \sigma^2)$	$\mathcal{N}(\mu, \tau^2)$	$\mathcal{N}(\rho(\sigma^2\mu + \tau^2x), \rho\sigma^2\tau^2)$ <i>where $\rho^{-1} = \sigma^2 + \tau^2$</i>
Poisson	$\mathcal{Poi}(\theta)$	$\mathcal{Ga}(\alpha, \eta)$	$\mathcal{Ga}(\alpha + x, \eta + 1)$
Gamma	$\mathcal{Ga}(\nu, \theta)$	$\mathcal{Ga}(\alpha, \eta)$	$\mathcal{Ga}(\alpha + \nu, \eta + x)$
Binomial	$\mathcal{Bin}(n, \theta)$	$\mathcal{Be}(\alpha, \eta)$	$\mathcal{Be}(\alpha + x, \eta + n - x)$

Table 1: Conjugate updates for standard exponential families.

Model Family	Likelihood $f(x \theta)$	Prior $\pi(\theta)$	Posterior $\pi(\theta x)$
Negative Binomial	$\mathcal{NB}(m, \theta)$	$\mathcal{Be}(\alpha, \eta)$	$\mathcal{Be}(\alpha + m, \eta + x)$
Multinomial	$\mathcal{M}_k(\theta_1, \dots, \theta_k)$	$\mathcal{Dir}(\alpha_1, \dots, \alpha_k)$	$\mathcal{Dir}(\alpha_1 + x_1, \dots, \alpha_k + x_k)$
Normal (unknown precision)	$\mathcal{N}(\mu, 1/\theta)$	$\mathcal{Ga}(\alpha, \eta)$	$\mathcal{Ga}\left(\alpha + 0.5, \eta + \frac{(\mu-x)^2}{2}\right)$

Table 2: Additional conjugate models and updates.

3 Lesson 3: Computational Approximations

Welcome to the first part of the Monte Carlo Simulation module. These notes will cover the foundations of computational statistics, specifically focusing on Crude Monte Carlo methods and Importance Sampling. Historically, exact analytical computation in statistics and probability can be mathematically demanding or virtually impossible. Consider the challenge of computing a relatively complex integral such as

$$\int_{(0,1)} e^{\theta^2 + \cos(\theta)} d\theta$$

The problem is even more apparent in combinatorial spaces. Let $\mathcal{S}_n$ represent the collection of all permutations of the set $\{1, \dots, n\}$. If we are tasked with evaluating the sum

$$\sum_{\sigma \in \mathcal{S}_n} e^{-\sum_{i=1}^n |\sigma_i - i|^2}$$

direct computation quickly becomes unmanageable. To understand why exact computation is not always feasible, imagine that evaluating each term $h(\sigma)$ takes just 0.001 seconds on a standard laptop. For $n = 15$ or $n = 20$, the number of permutations is $n! = 1 \cdot 2 \cdot \dots \cdot n$, meaning the time required to compute the entire sum would span an impractical number of years. Consequently, we must rely on approximation techniques.

3.1 The Monte Carlo Principle

The core problem in many statistical applications is the need to compute an integral of the form

$$\mathfrak{J} = \mathbb{E}_{\pi}[h(\theta)] = \int_{\Theta} h(\theta) \pi(\theta) d\theta$$

Here, Θ is a subset of $\mathbb{R}^d$, π represents a probability density, and h is a given function. In the context of Bayesian statistics, π is typically the posterior density, denoted as $\pi(\cdot|x)$. The notation $\mathbb{E}_{\pi}$ emphasizes that the mathematical expectation is taken with respect to the probability distribution π .

Example: Bayesian Inference

Suppose we have a subset $\Theta_0 \subset \Theta$ and we want to test its posterior probability. We can express this as:

$$\pi(\Theta_0|x) = \int_{\Theta_0} \pi(\theta|x) d\theta = \int_{\Theta} h(\theta) \pi(\theta|x) d\theta$$

In this scenario, $h(\theta)$ acts as an indicator function, equal to 1 if $\theta \in \Theta_0$ and 0 otherwise.

Alternatively, if we wish to compute the posterior mean of θ , our function simply becomes $h(\theta) = \theta$. The integral takes the form:

$$E_{\pi}[\theta|x] = \int_{\Theta} \theta \pi(\theta|x) d\theta$$

To approximate these integrals, we rely on the Monte Carlo Principle. By extracting a sample $(\theta_1, \dots, \theta_m)$ from the density π , we can approximate the integral $\mathfrak{J}$ using the empirical average:

$$\bar{h}_m = \frac{1}{m} \sum_{j=1}^m h(\theta_j)$$

Theorem & Rules: Strong Law of Large Numbers (SLLN)

The theoretical justification for this method is rooted in the SLLN. If the generated samples θ_j are independent and identically distributed (i.i.d.) following the law π , then the empirical average converges to the true expected value:

$$\bar{h}_m = \frac{1}{m} \sum_{j=1}^m h(\theta_j) \longrightarrow \mathbb{E}_\pi[h(\theta)] = \int h(\theta)\pi(\theta)d\theta$$

with probability one.

For instance, to approximate our introductory integral $\int_0^1 e^{\theta^2 + \cos(\theta)} d\theta$, we can define $\pi(\theta) = 1$ on the interval $(0, 1)$ —representing the standard uniform distribution—and set our function as $h(\theta) = e^{\theta^2 + \cos(\theta)}$.

ALGORITHM: BASIC MONTE CARLO METHOD

For $i = 1, \dots, m$,

simulate $\theta_i \sim \pi(\theta)$.

Take

$$\hat{\mathfrak{J}}_m = (h(\theta_1) + \dots + h(\theta_m))/m$$

to approximate $\mathfrak{J}$.

As long as the (computer-generated) pseudo-random generation from g is feasible and the values $h(\xi)$ are computable. When simulation from g is a problem because g is nonstandard and usual techniques such as accept-reject algorithms (not covered here) are difficult to devise, more advanced techniques such as Markov Chain Monte Carlo (MCMC) are required.

Obviously, this *Monte Carlo estimate* of $\mathfrak{J}$ is not exact, but generating a sufficiently large number of random variables can render this approximation error arbitrarily small in a suitable probabilistic sense. It is even possible to assess the size of this error. If

$$\int |h(x)|^2 g(x) dx < \infty,$$

the central limit theorem shows that $\sqrt{n} [\hat{\mathfrak{J}}_n - \mathfrak{J}]$ is also normally distributed, and this can be used to construct asymptotic confidence regions for $\hat{\mathfrak{J}}_n$, estimating the asymptotic variance from the simulation output.

3.2 Data Augmentation and Mixture of Densities

Direct sampling from complex target distributions can be tricky. Consider a scenario where we want to simulate from a mixture of two Beta densities:

$$\pi(\theta) = 0.4\pi_1(\theta) + 0.6\pi_2(\theta)$$

where $\pi_1 = \text{Beta}(2, 8)$ and $\pi_2 = \text{Beta}(10, 2)$.

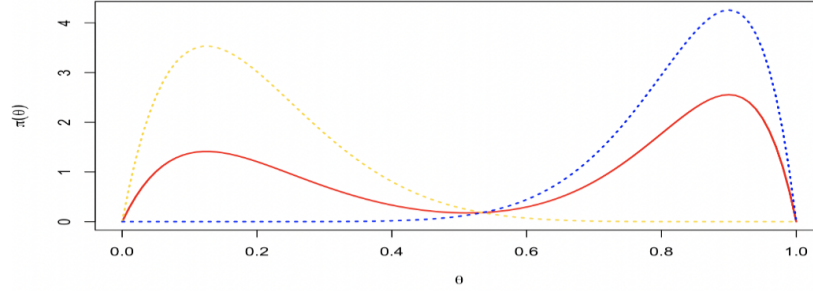


Figure 7: Mixture of two Beta densities

Important Note

To efficiently sample from this mixture, we utilize a technique called **Data Augmentation**. We can augment the variable θ by introducing a hierarchical indicator variable $I \in \{1, 2\}$. We define the probabilities $P(I = 1) = p_1$ and $P(I = 2) = p_2$, ensuring $p_1 + p_2 = 1$. The conditional distributions become $P(\theta|I = 1) \sim \text{Beta}(a_1, b_1)$ and $P(\theta|I = 2) \sim \text{Beta}(a_2, b_2)$. By formulating the joint density $P(I = i, \theta)$ and marginalizing over I , we perfectly recover the target mixture density $\pi(\theta)$.

The algorithm for sampling this mixture proceeds iteratively for $t = 1, \dots$:

1. Sample the indicator $I^{(t)}$ from $\{1, 2\}$ with probabilities corresponding to the mixture weights.
2. Given the sampled state $I^{(t)} = i$, draw $\theta^{(t)}$ from the corresponding conditional density $\text{Beta}(a_i, b_i)$.

By ignoring the instrumental variable I , the resulting sequence of $\theta^{(t)}$ forms an i.i.d. sample from the desired target mixture π .

3.3 Changing the Density: A Practical Example

Sometimes, we can dramatically simplify our calculations by changing the density from which we simulate. Consider estimating a normal mean using a robust Cauchy prior: $x \sim \mathcal{N}(\theta, 1)$ and

$\theta \sim \mathcal{C}(0, 1)$. Under a squared error loss function, the posterior mean is calculated as:

$$\delta^\pi(x) = \frac{\int_{-\infty}^{\infty} \frac{\theta}{1+\theta^2} e^{-(x-\theta)^2/2} d\theta}{\int_{-\infty}^{\infty} \frac{1}{1+\theta^2} e^{-(x-\theta)^2/2} d\theta}$$

Instead of attempting to simulate from this complex posterior, we can re-express the numerator and denominator as expectations with respect to a Normal distribution $\mathcal{N}(x, 1)$. This suggests we can simulate i.i.d. variables $\theta_1, \dots, \theta_m \sim \mathcal{N}(x, 1)$ and calculate the empirical estimator:

$$\hat{\delta}_m^\pi(x) = \frac{\sum_{i=1}^m \frac{\theta_i}{1+\theta_i^2}}{\sum_{i=1}^m \frac{1}{1+\theta_i^2}}$$

Thanks to the Strong Law of Large Numbers, the numerator converges to the integral I_1 , the denominator converges to I_2 , and the overall estimator $\hat{\delta}_m^\pi(x)$ smoothly converges to the true posterior mean $\delta^\pi(x)$ as $m \rightarrow \infty$.

3.4 Importance Sampling

When direct sampling from the target distribution π is highly inefficient or structurally impossible, **Importance Sampling** serves as a powerful alternative.

Definition: Importance Sampling Representation

We can rewrite our target integral using an instrumental (or proposal) distribution μ :

$$\mathbb{E}_\pi[h(\theta)] = \int_{\mathcal{X}} \left[h(\theta) \frac{\pi(\theta)}{\mu(\theta)} \right] \mu(\theta) d\theta = \mathbb{E}_\mu \left[h(\theta) \frac{\pi(\theta)}{\mu(\theta)} \right]$$

This alternative representation allows us to compute expectations with respect to μ instead of π .

The algorithm requires generating a sample $\theta_1, \dots, \theta_m$ from the instrumental distribution μ . We then approximate the integral using a weighted average:

$$\frac{1}{m} \sum_{j=1}^m w(\theta_j) h(\theta_j)$$

The term $w(\theta_j) := \frac{\pi(\theta_j)}{\mu(\theta_j)}$ is known as the *importance weight*.

Theorem & Rules: Convergence of the Estimator

The estimator converges to $\mathbb{E}_\pi[h(\theta)]$ for any choice of the instrumental distribution μ , provided that the support of μ fully contains the support of π , mathematically expressed as $\text{supp}(\mu) \supset \text{supp}(\pi)$.

A major advantage is that the same sample generated from μ can be reused repeatedly to evaluate different functions h or even different target densities π .

Important Note

In many practical scenarios, the densities π and μ are only known up to a normalizing constant. In these cases, we use the self-normalized alternative:

$$\frac{\sum_{j=1}^m h(\theta_j) \frac{\pi(\theta_j)}{\mu(\theta_j)}}{\sum_{j=1}^m \frac{\pi(\theta_j)}{\mu(\theta_j)}}$$

While this estimator introduces a small bias, it still converges to $\mathbb{E}_\pi[h(\theta)]$ by the SLLN and can frequently outperform the unbiased estimator when evaluated under squared error loss.

ALGORITHM: IMPORTANCE SAMPLING METHOD

For $i = 1, \dots, m$,

 simulate $\theta_i \sim \mu(\theta)$;

 compute $\omega_i = \pi(\theta_i)/\mu(\theta_i)$.

Take

$$\hat{\mathfrak{J}}_m = \sum_{i=1}^m \omega_i h(\theta_i) \Big/ \sum_{i=1}^m \omega_i$$

to approximate $\mathfrak{J}$.

4 Lesson 4: Markov Chains and MCMC Methods

These notes present the second major segment of our study on Monte Carlo Simulations. We will be expanding upon foundational statistical computing concepts by introducing advanced algorithms that are indispensable in modern computational statistics. The primary topics covered in this module include Markov Chains and the Ergodic theorem, the broader framework of Markov Chain Monte Carlo (MCMC) methods, the Hastings-Metropolis algorithm, and the Gibbs sampling algorithm.

4.1 Markov Chain Monte Carlo (MCMC) Methods

When navigating the complexities of most models encountered in Bayesian modeling, traditional Crude Monte Carlo methods are typically not feasible. Because of these computational limitations in high-dimensional or analytically difficult spaces, MCMC techniques have become the fundamental core of Bayesian computing.

4.1.1 The Core Idea Behind MCMC

The foundational idea of MCMC is to simulate an intelligent sequence of values from a Markov chain, denoted as $(\theta^{(t)})_{t \in \mathbb{N}}$.

The primary objective is to design this sequence such that, in the long run, the values $\theta^{(t)}$ behave precisely as if they were independently sampled from our target distribution, $\pi(\cdot)$. To accomplish this, the algorithm must construct the chain in a specific way ensuring that $\pi(\cdot)$ acts as its stationary distribution. It is important to remember that in standard Bayesian applications, this target π represents the posterior density, denoted as $\pi(\cdot|x)$.

4.2 Understanding Markov Chains

To understand MCMC, we must first establish a formal understanding of Markov chains themselves.

Definition: Homogeneous Markov Chain

A stochastic process $(\theta^{(t)})_{t \in \mathbb{N}}$ is defined as a homogeneous Markov chain if the conditional distribution of the current state $\theta^{(t)}$, given its entire past history $(\theta^{(0)}, \dots, \theta^{(t-1)})$, depends exclusively on the immediately preceding state $\theta^{(t-1)}$. Furthermore, this transition mechanism remains the same for all $t \geq 1$.

Formally, for a sequence of random variables $\theta^0, \theta^1, \dots, \theta^n, \dots$, the Markov property is expressed as:

$$P(\theta^{t+1} = \theta | \theta^0, \theta^1, \theta^2, \dots, \theta^t) = P(\theta^{t+1} = \theta | \theta^t) = K(\theta^t, \theta)$$

Here, the function K is referred to as the *transition kernel* of the chain.

Important Note

Be very careful with informal notation. While we frequently write $P(\theta^{t+1} = \theta | \theta^t)$, if the conditional distribution of θ^{t+1} is continuous, this expression must strictly be interpreted as the conditional density function of θ^{t+1} given θ^t , evaluated exactly at the point θ .

4.2.1 Standard Notation

The transition kernel is the engine of the Markov chain. It defines the probability of moving from an old state x to a new state y . This is formally noted as:

$$K(x, y) = P\{\theta^t \in dy | \theta^{t-1} = x\}$$

In the literature, you may encounter several alternative notations for this transition, including $K(x, y)$, $K(x \rightarrow y)$, or the conditional format $K(y|x)$.

4.3 Invariant Distributions & Convergence

For MCMC to work, our generated chain must eventually settle into the target distribution. This concept is formalized through invariant distributions and the principle of ergodicity.

Definition: Invariant Distribution

A transition kernel K is said to have an invariant (or stationary) distribution π if it satisfies the following integral equation:

$$\int K(\theta, B) \pi(\theta) d\theta = \int_B \pi(\theta) d\theta$$

To interpret this practically: suppose at step k , the state θ^k is drawn from the distribution π . If the next state θ^{k+1} is generated conditional on θ^k using our kernel (i.e., $\theta^{k+1} | \theta^k \sim K(\cdot | \theta^k)$), then the resulting marginal distribution of θ^{k+1} will also be exactly π .

Important Note

While the property of invariance guarantees that both θ^k and θ^{k+1} possess the exact same marginal distribution π , it is crucial to recognize that they are **not** independent random variables.

4.3.1 Ergodicity and the Law of Large Numbers

The ultimate utility of MCMC relies on a specific form of the Law of Large Numbers adapted for Markov chains.

Definition: Ergodicity

Informally, a Markov chain is considered ergodic if, after running the chain for a sufficiently large number of steps t , the state $\theta^{(t)}$ is approximately distributed according to the target $\pi(\theta)$. Crucially, this convergence must hold true regardless of the arbitrary starting value $\theta^{(0)}$ chosen to initialize the chain.

Theorem & Rules: Strong Law of Large Numbers for Markov Chains

If the constructed chain $(\theta^{(t)})$ is ergodic and features π as its stationary distribution, then the empirical average of any function g evaluated over the chain's path will converge almost surely to the expected value of g under π .

Mathematically, with probability one:

$$\frac{1}{T} \sum_{t=1}^T g(\theta^{(t)}) \rightarrow \mathbb{E}_{\pi}[g(\theta)] = \int g(\theta) \pi(d\theta)$$

4.4 Implementing MCMC Algorithms

The foundational methodology of any MCMC approach involves a few straightforward, yet powerful, steps. The primary objective is to find a transition kernel K that possesses our target distribution π as its stationary distribution. Once an appropriate kernel is identified, we simulate a Markov chain $(\theta^{(t)})_{t \geq 1}$ from K .

With this simulated chain, we can approximate the expected value $\mathbb{E}_{\pi}[g(\theta)]$ using the empirical average, defined as $\hat{g}_T := \frac{1}{T} \sum_{t=1}^T g(\theta^{(t)})$. If an algorithm successfully generates such a chain, the ergodic theorem guarantees that, in almost all settings, the empirical average $\hat{g}_T$ will converge to $\mathbb{E}_{\pi}[g(\theta)]$ for almost any starting value.

Important Note

While the ergodic theorem provides strong guarantees, certain mathematical conditions (such as irreducibility) are required to ensure ergodicity. Furthermore, even when $\theta^{(t)}$ is approximately distributed from π for large t , one must be careful to account for the inherent correlations between the sequential draws of $\theta^{(t)}$. The main practical issue in Bayesian computing remains: how exactly can we produce a Markov Chain with a given stationary distribution π (or a posterior $\pi(\cdot|x)$)?

4.5 The Metropolis-Hastings Algorithm

The Metropolis-Hastings (MH) algorithm provides a flexible and universal solution to construct the required Markov chain. Let π be our desired target distribution. To navigate the state space, we must first choose a transition kernel $q(\cdot|\cdot)$, which serves as our proposal distribution.

4.5.1 Algorithm Steps

The Metropolis-Hastings algorithm proceeds as follows:

1. **Initialization:** Start with an arbitrary initial value $\theta^{(0)} = \theta_0$.
2. **Proposal generation:** At the t -th iteration, given the current state $\theta^{(t)}$, generate a candidate state $\theta^{(*)}$ from the proposal distribution: $\theta^{(*)} \sim q(\theta^{(*)}|\theta^{(t)})$.
3. **Acceptance probability:** Compute the probability of accepting this new candidate. This is calculated as:

$$\alpha(\theta^{(t)}, \theta^{(*)}) = \min \left\{ \frac{\pi(\theta^{(*)}) q(\theta^{(t)}|\theta^{(*)})}{\pi(\theta^{(t)}) q(\theta^{(*)}|\theta^{(t)})}, 1 \right\}$$

4. **State transition:**

- With probability $\alpha(\theta^{(t)}, \theta^{(*)})$, accept the move and set $\theta^{(t+1)} = \theta^{(*)}$.
- With probability $1 - \alpha(\theta^{(t)}, \theta^{(*)})$, reject the move and remain at the current state, setting $\theta^{(t+1)} = \theta^{(t)}$.

The sequence $(\theta^{(t)})_{t \geq 1}$ generated by this algorithm yields a Markov chain whose stationary distribution is exactly π . A massive advantage of this algorithm is that to run it, one only needs to know π up to a normalizing constant. Because the algorithm relies on the ratio $\frac{\pi(\theta_2) q(\theta_1|\theta_2)}{\pi(\theta_1) q(\theta_2|\theta_1)}$, any intractable constants in the target density cancel out seamlessly.

4.5.2 Theoretical Justification: Detailed Balance (Optional)

Why does the Metropolis-Hastings algorithm work so well? The rationale is grounded in the concept of reversibility and the detailed balance condition.

Definition: Detailed Balance Condition

A Markov chain with transition kernel K satisfies the detailed balance condition if there exists a function q that satisfies the equation:

$$K(\theta^2|\theta^1)q(\theta^1) = K(\theta^1|\theta^2)q(\theta^2)$$

for every pair of states (θ^1, θ^2) .

Theorem & Rules: Reversibility and Invariance

Suppose that a Markov chain with transition function K satisfies the detailed balance condition with a probability density function $q = \pi$. Then, the density π is the invariant (stationary) density of the chain.

Thus, we can confidently build a chain targeting $\pi(\cdot)$ by finding a suitable reversible kernel K satisfying $K(\theta^2|\theta^1)\pi(\theta^1) = K(\theta^1|\theta^2)\pi(\theta^2)$. The MH algorithm guarantees convergence to π precisely

because its transition kernel structurally satisfies this detailed balance condition. Mathematical proofs confirm that $K(\theta_{new}|\theta_{old})\pi(\theta_{old})$ equates identically to $K(\theta_{old}|\theta_{new})\pi(\theta_{new})$.

4.5.3 Variations of Metropolis-Hastings

By tweaking the proposal distribution q , we can derive various specialized forms of the MH algorithm.

4.5.4 Independent Metropolis-Hastings

The Independent-MH variation assumes a proposal kernel $q(\theta_{new}|\theta_{old}) = f(\theta_{new})$, where f is a standalone density entirely independent from the previous state θ_{old} .

In this setup, given $\theta^{(t)}$, we sample our candidate θ_* directly from $f(\theta_*)$. We then set $\theta^{(t+1)} = \theta_*$ with the modified acceptance probability $\min\left(\frac{\pi(\theta_*)f(\theta^{(t)})}{\pi(\theta^{(t)})f(\theta_*)}, 1\right)$. Otherwise, the state remains $\theta^{(t+1)} = \theta^{(t)}$.

4.5.5 Random Walk Metropolis-Hastings

The Random-Walk MH (RW-MH) algorithm utilizes a symmetric proposal distribution, meaning $q(\theta_1|\theta_2) = q(\theta_2|\theta_1)$. Typically, this is formulated as $q(\theta_{new}|\theta_{old}) = f(\theta_{new} - \theta_{old})$, where f is a symmetric density function.

Because the proposal is symmetric, the proposal ratio perfectly cancels out, drastically simplifying the acceptance probability to $\alpha(\theta_1, \theta_2) = \min\left\{\frac{\pi(\theta_2)}{\pi(\theta_1)}, 1\right\}$. During execution, we sample θ_* from $f(\cdot - \theta^{(t)})$ and accept it as the new state with probability $\min\left(\frac{\pi(\theta_*)}{\pi(\theta^{(t)})}, 1\right)$.

Example: Gaussian Random Walk

A highly practical and important application of RW-MH in continuous spaces ($\Theta = \mathbb{R}^k$) is the Gaussian RW-MH. Here, the function f is chosen as a normal density characterized by a specific covariance matrix Σ_W .

Operationally, the proposed parameter θ_* is generated by adding Gaussian noise to the old state:

$$\theta_* = \theta_{old} + \epsilon$$

where ϵ is a Gaussian random vector with covariance matrix Σ_W .

4.5.6 Heuristics of the Symmetric Kernel

To better understand the mechanics of the Random-Walk Metropolis-Hastings (RW-MH) algorithm, consider the heuristics behind its acceptance probability. For a symmetric kernel, the acceptance probability simplifies to $\alpha = \min\left(\frac{\pi(\theta_*)}{\pi(\theta^{(t)})}, 1\right)$.

- If the candidate state θ_* lies in a region of higher or equal probability compared to the current state (i.e., $\pi(\theta_*) \geq \pi(\theta^{(t)})$), the chain will accept the jump to θ_* with a probability of exactly 1.

- Conversely, if the candidate state represents a move to a region of lower probability ($\pi(\theta_*) < \pi(\theta^{(t)})$), the chain will still jump to θ_* , but only with a probability $\alpha = \pi(\theta_*)/\pi(\theta^{(t)})$.

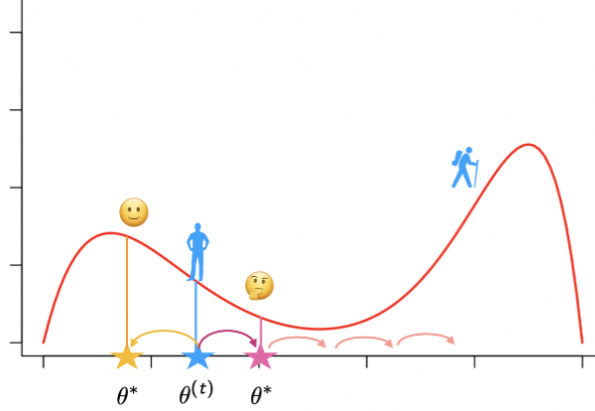


Figure 8: Acceptance probability as a function of the ratio $\pi(\theta_*)/\pi(\theta^{(t)})$

4.6 Choosing the Proposal Distribution

While the theoretical guarantees of convergence for the Metropolis-Hastings algorithm are very high, the practical success of the simulation relies heavily on the choice of the proposal distribution, q .

4.6.1 Consequences of a Poor Proposal

An inappropriate selection for q can severely compromise the efficiency of the MCMC sampler. Two primary issues can arise:

- **High Rejection Rates:** If the proposal distribution dictates unrealistic jumps, the chain may suffer from very high rejection rates. In this scenario, the chain hardly moves, resulting in a highly inefficient exploration of the state space.
- **Myopic Exploration:** Alternatively, if the proposed jumps are too small, the chain might exhibit a myopic exploration of the support of π . This causes a heavy dependence on the starting value $\theta^{(0)}$, with the chain becoming stuck in a local neighborhood mode close to $\theta^{(0)}$.

4.6.2 Requirements for a Robust Proposal

To guarantee a comprehensive exploration of the state space, the proposal distribution q needs to be strictly positive everywhere on the support of the target density π . Furthermore, for optimal exploration of this support, the ratio π/q needs to be bounded. If it is not bounded, the Markov chain will take an unacceptably long time to reach regions where the ratio q/π exhibits low values.

Example: Tuning the Random Walk Sampler

The Gaussian Random Walk (GRW) provides considerable flexibility. One can modify its tails—for instance, using a Student’s t-distribution instead of a Normal distribution for heavier tails—or adjust its scale to control the size of the neighborhood explored. Practically, one common tuning strategy during the initial burn-in stage is to calibrate the proposal to achieve an acceptance probability of roughly 0.234, often affectionately referred to as the “magic number” in MCMC literature.

4.7 Convergence Assessment and Diagnostics

A capital question in MCMC simulations is: *How many iterations do we need to run?* Evaluating whether a chain has properly converged to its stationary distribution requires careful statistical diagnostics.

4.7.1 General Rules of Thumb

Theorem & Rules: Rules for Convergence

- **Rule 1:** There is no absolute, universal number of simulations required; 1,000 iterations, for example, is intrinsically neither large nor small.
- **Rule 2:** It typically takes much longer to check for convergence than for the chain itself to actually reach convergence.
- **Rule 3:** MCMC is inherently a “what-you-get-is-what-you-see” algorithm, meaning it fails to tell you about unexplored parts of the target space.
- **Rule 4:** When in doubt about convergence, run multiple MCMC chains in parallel and check them for consistency.

While there are many “quick-and-dirty” solutions available in the literature to assess this, they are not necessarily trustworthy on their own.

4.7.2 Key Diagnostic Metrics

To assess the health of the chain, practitioners should regularly check a few fundamental metrics:

- **Acceptance Rate:** This is simply the number of accepted moves divided by the total number of moves. As a general rule of thumb, if the acceptance rate is too high or too small, it denotes some kind of problem with the simulation design.
- **Trace Plots:** Visualizing the path of the chain over time. If these plots appear too regular (e.g., highly periodic), there is likely a structural problem. Additionally, if trace plots look drastically different depending on the starting point, the same conclusion applies.

- **Auto Correlation Function (ACF):** Evaluates the correlation between states at different lag intervals within the generated chain.

4.8 Burn-in, Thinning, and Effective Sample Size

Two extra parameters are frequently employed to manage chain correlation and dependence on initial conditions:

Important Note

Burn-in: This involves starting to use the chain in the estimation procedure only after discarding the first part of the chain. This practice is highly useful in forgetting the starting point and helping the overall mixing. If the burn-in period is designated as M_0 , the resulting estimator utilizes the sequence: $\frac{1}{M-M_0} \sum_{j=M_0+1}^M h(\theta^{(j)})$.

Thinning: To actively reduce correlation, one can use only a sub-sequence by thinning the original chain. The thinned sequence becomes $\theta^{(1)}, \theta^{(1+\delta)}, \theta^{(1+2\delta)}, \dots, \theta^{(1+k\delta)}$. The resulting estimator under thinning is $\frac{1}{K} \sum_{j=1}^K h(\theta^{(M_0+j\delta)})$.

Definition: Effective Sample Size

Because the states in an MCMC output are correlated, N simulations from the algorithm are not entirely equivalent to independent i.i.d. simulations. The Auto-Correlation (ACF) at lag k is formally defined as $\rho_k = \text{corr}(x^{(t)}, x^{(t+k)})$.

The *Effective Sample Size* (ESS), based on the estimated k -th order auto-correlations $\hat{\rho}_k$, is defined as:

$$N^{ess} = N \left(1 + 2 \sum_{k=1}^{T_0} \hat{\rho}_k^2 \right)^{-1/2}$$

It is crucial to note that the ESS is only a partial indicator; it completely fails to signal situations where chains are permanently stuck in one single mode of the target distribution.

4.9 The Gibbs Sampler

The second foundational algorithm in our MCMC toolkit is the Gibbs sampler. Unlike the Metropolis-Hastings algorithm which proposes moves in the full parameter space, the Gibbs sampler takes advantage of the inherent structural properties of the target distribution.

Specifically, if our goal is to simulate from a marginal distribution such as:

$$\pi(\theta_1) = \int \pi_1(\theta_1|\theta_2)\pi_2(\theta_2)d\theta_2$$

we recognize that simulating from $\pi(\theta_1)$ naturally comes from simulating from the joint distribution:

$$\pi_1(\theta_1|\theta_2)\pi_2(\theta_2)$$

Example: Beta Mixture Model

Recall the mixture example. In the Beta mixture scenario discussed previously, one can define a latent variable $\theta_2 = I$, which takes values of 1 or 2 with probabilities p and $1 - p$, respectively. This structural decomposition allows us to sample the parameters conditionally rather than directly from the complex mixture.

4.9.1 The Two-Stage Gibbs Sampler

Let us focus on the standard Bayesian case in which the target distribution we wish to sample from is a bivariate posterior, denoted as $\pi(\theta_1, \theta_2|x)$.

Typically, the marginal posterior $\pi_2(\theta_2|x)$ is analytically unavailable or difficult to simulate directly. However, it frequently happens that the conditional distributions—specifically $\pi_1(\theta_1|x, \theta_2)$ and $\pi_2(\theta_2|x, \theta_1)$ —are known and can be easily simulated.

Important Note

The core idea is to create a Markov chain based entirely on these available conditional distributions. One must note, mathematically, that the product of these conditionals does not equal the joint distribution:

$$\pi_1(\theta_1|x, \theta_2)\pi_2(\theta_2|x, \theta_1) \neq \pi_1(\theta_1|x, \theta_2)\pi_2(\theta_2|x) = \pi(\theta_1, \theta_2|x)$$

Despite this inequality, sequentially sampling from them works beautifully. As the Romans said, *fortuna audaces iuvat* (fortune favors the brave).

Theorem & Rules: Two-Stage Gibbs Algorithm

The algorithm sequentially updates each parameter by sampling from its full-conditional distribution.

1. **Initialization:** Start with an arbitrary starting value $\theta_2^{(0)}$.
2. **Iteration t :** Given the previous state $\theta_2^{(t-1)}$, generate the new states by sequentially sampling:
 - $\theta_1^{(t)}$ according to the conditional $\pi_1(\theta_1|x, \theta_2^{(t-1)})$.
 - $\theta_2^{(t)}$ according to the conditional $\pi_2(\theta_2|x, \theta_1^{(t)})$.

Theorem: The target distribution $\pi(\theta_1, \theta_2|x)$ is exactly the stationary distribution for this transition mechanism.

4.9.2 Proof of Invariance (Optional)

To understand why the Gibbs sampler is valid, we can compute the transition kernel of the Gibbs chain $(\theta_1^t, \theta_2^t)_{t \geq 0}$. The kernel for transitioning from an old state to a new state is formally defined

as:

$$K(\theta_{1,new}, \theta_{2,new} | \theta_{1,old}, \theta_{2,old}) = \pi_1(\theta_{1,new} | x, \theta_{2,old}) \pi_2(\theta_{2,new} | x, \theta_{1,new})$$

To prove that this kernel K has the invariant distribution π , we need to mathematically show that:

$$\int \int K(\theta_{1,new}, \theta_{2,new} | \theta_{1,old}, \theta_{2,old}) \pi(\theta_{1,old}, \theta_{2,old} | x) d\theta_{1,old} d\theta_{2,old} = \pi(\theta_{1,new}, \theta_{2,new} | x)$$

Using the standard probability factorization rule, we can rewrite the joint distribution of the old state as:

$$\pi(\theta_{1,old}, \theta_{2,old} | x) = \pi_1(\theta_{1,old} | \theta_{2,old}, x) \pi_2(\theta_{2,old} | x)$$

Substituting the kernel and the factored joint distribution into our integral yields:

$$\begin{aligned} &= \int \int \pi_1(\theta_{1,new} | x, \theta_{2,old}) \pi_2(\theta_{2,new} | x, \theta_{1,new}) \pi(\theta_{1,old}, \theta_{2,old} | x) d\theta_{1,old} d\theta_{2,old} \\ &= \int \pi_1(\theta_{1,new} | x, \theta_{2,old}) \left(\int \pi_1(\theta_{1,old} | \theta_{2,old}, x) d\theta_{1,old} \right) \pi_2(\theta_{2,old} | x) \pi_2(\theta_{2,new} | x, \theta_{1,new}) d\theta_{2,old} \end{aligned}$$

Since the inner integral over the conditional density π_1 evaluates to 1, we are left with:

$$= \pi_2(\theta_{2,new} | x, \theta_{1,new}) \int \pi_1(\theta_{1,new} | x, \theta_{2,old}) \pi_2(\theta_{2,old} | x) d\theta_{2,old}$$

The remaining integral is exactly the marginalization that results in the marginal posterior of $\theta_{1,new}$. Thus:

$$= \pi_2(\theta_{2,new} | x, \theta_{1,new}) \pi_1(\theta_{1,new} | x) = \pi(\theta_{1,new}, \theta_{2,new} | x)$$

This concludes the proof.

4.9.3 Convergence and Implementation Details

This sampling procedure produces a Markov Chain $(\theta^{(t)})_{t \geq 1}$ with the exact stationary distribution $\pi(\theta | x)$. Because of this, we can rely on the Ergodic theorem to approximate posterior expectations.

Specifically, the expected value:

$$\int h(\theta) \pi(\theta | x) d\theta$$

can be reliably approximated by the empirical average:

$$\bar{h}_M = \frac{1}{M} \sum_{j=1}^M h(\theta^{(j)})$$

This holds true even though the samples $\theta^{(j)}$ generated by the Gibbs sampler are not strictly independent and identically distributed (i.i.d.) from $\pi(\theta | x)$. In the long run, the states $\theta^{(t)} \rightarrow \theta \sim \pi(\cdot | x)$.

Important Note

Implementation Guidelines To effectively implement a Gibbs sampler, you must carefully derive an efficient decomposition of the joint target distribution into conditionals that are easy to simulate from. In many complex models, it may be highly useful—or even necessary—to add latent (unobserved) variables to achieve a simulable set of full-conditionals.

4.10 A Practical Example of Gibbs Sampling

To solidify our understanding of the Gibbs sampler, let us walk through a practical implementation using a continuous Bayesian hierarchical model.

4.10.1 Model Specification

Consider a scenario where we have n observations x_i drawn from a normal distribution with an unknown mean μ and unknown variance σ^2 . Instead of working directly with the variance, it is often mathematically convenient to work with the precision, denoted as $\tau = \sigma^{-2}$.

The model relies on the following priors and likelihood structure:

- **Likelihood:** $x_i \sim \mathcal{N}(\mu, \sigma^2)$
- **Prior on Precision:** $\tau = \sigma^{-2} \sim \mathcal{G}(\alpha_0, \text{rate} = \beta_0)$
- **Prior on Mean:** $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$

4.10.2 Deriving the Posterior and Full Conditionals

To implement the Gibbs sampler, we must first establish the joint likelihood of the data and the joint prior. The likelihood function of the data given the parameters is:

$$f(x_{1:n}|\mu, \tau) = \frac{\tau^{n/2}}{(2\pi)^{n/2}} e^{-\frac{\tau}{2} \sum_{i=1}^n (x_i - \mu)^2}$$

The joint prior distribution for the parameters τ and μ is:

$$\pi(\tau, \mu) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left\{-\frac{1}{2\sigma_0^2}(\mu - \mu_0)^2\right\} \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} e^{-\beta_0\tau} \tau^{\alpha_0-1}$$

By multiplying the likelihood by the prior, we obtain the unnormalized posterior density:

$$\pi(\tau, \mu|x_{1:n}) \propto \tau^{\frac{n}{2} + \alpha_0 - 1} e^{-\frac{1}{2\sigma_0^2}(\mu - \mu_0)^2 - \beta_0\tau - \frac{\tau}{2} \sum_{i=1}^n (x_i - \mu)^2}$$

Important Note

It is crucial to note that this specific hierarchical model is not fully conjugate. We cannot directly recognize the joint posterior as a standard, easily simulable distribution. However, we can overcome this by deriving the full-conditional distributions for each parameter individually.

Example: Full-Conditional Distributions

Despite the lack of joint conjugacy, the individual full-conditionals take the form of well-known distributions:

1. The conditional distribution for the mean: By dropping all terms in the posterior that do not depend on μ , we find that the conditional distribution is Normal:

$$\mu|\sigma, x \sim \mathcal{N}(\mu_n, \sigma_n^2)$$

where the updated parameters are defined as:

$$\mu_n = \frac{\sigma^2}{\sigma^2 + n\sigma_0^2}\mu_0 + \frac{n\sigma_0^2}{\sigma^2 + n\sigma_0^2} \frac{1}{n} \sum_{i=1}^n x_i, \quad \sigma_n^2 = \frac{\sigma^2\sigma_0^2}{\sigma^2 + n\sigma_0^2}$$

2. The conditional distribution for the precision: Similarly, by isolating terms dependent on τ (where $\tau = 1/\sigma^2$), the resulting conditional is a Gamma distribution:

$$1/\sigma^2|\mu, x \sim \mathcal{G}(a_n, b_n)$$

where the updated shape and rate parameters are:

$$a_n = \alpha_0 + n/2$$
$$b_n = \beta_0 + \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2$$

Note: Practical implementations of this model, including iterative sampling and diagnostic checking, are traditionally handled via statistical software such as R.

4.11 The General Gibbs Sampler

Having seen a two-parameter example, we can now generalize the Gibbs sampling algorithm for models with an arbitrary number of parameters.

Let $\theta = (\theta_1, \dots, \theta_p)$ represent the full vector of parameters. We define $\theta_{-i} = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots)$ as the vector containing all parameters *except* θ_i . For any given joint distribution $\pi(\theta|x)$, the full conditionals are the univariate conditional distributions denoted by $\pi_i(\theta_i|x, \theta_{-i})$.

Theorem & Rules: General Gibbs Sampler Algorithm

At each iteration $t = 1, \dots, M$, given the current state of the parameter vector $(\theta_1^{(t)}, \dots, \theta_p^{(t)})$, we simulate the next state by sequentially updating each parameter using the most recently generated values. The sequential updates proceed as follows:

1. Sample $\theta_1^{(t+1)} \sim \pi_1(\theta_1 | \theta_2^{(t)}, \dots, \theta_p^{(t)}, x)$
2. Sample $\theta_2^{(t+1)} \sim \pi_2(\theta_2 | \theta_1^{(t+1)}, \theta_3^{(t)}, \dots, \theta_p^{(t)}, x)$
- ...
- P. Sample $\theta_p^{(t+1)} \sim \pi_p(\theta_p | \theta_1^{(t+1)}, \dots, \theta_{p-1}^{(t+1)}, x)$

By continually cycling through these conditional updates, the simulated parameter vectors will form a Markov chain whose stationary distribution is exactly the target joint posterior $\pi(\theta|x)$.

5 Lesson 5: Introduction to Regression

A large fraction of statistical analyses deals with representation of dependences between several variables. Linear regression is one of the most widespread tools of Statistics for analysing the (linear) influence of some variables or some factors on others. Understanding these foundations is crucial for applying statistical modeling to real-world data effectively.

5.1 Motivating Examples

Example: Bodyfat Estimation

The primary objective of this dataset is to model the percent body fat derived from Siri's (1956) equation, which serves as our response variable (y). The rationale for this modeling is that accurate measurement of body fat is inconvenient and costly. Therefore, it is desirable to have easy methods of estimating body fat that are not inconvenient or costly. The predictive model utilizes physical measurements:

- x_1 : Weight of the individual in pounds.
- x_2 : Chest circumference measured in cm.
- x_3 : Wrist circumference measured in cm.

This dataset is accessible from the BAS package in R.

Example: Pine Processionary Caterpillars

In this ecological study, the outcome variable of interest is the Pine processionary caterpillar colony size (y). This colony size is influenced by multiple environmental covariates:

- x_1 : Altitude of the area.
- x_2 : Slope measured in degrees.
- x_3 : Number of pines in the area.
- x_4 : Height of the central tree.
- x_5 : Diameter of the central tree.
- x_6 : Index of the settlement density.
- x_7 : Orientation of the area (ranging from 1 [southbound] to 2).
- x_8 : Height of the dominant tree.
- x_9 : Number of vegetation strata.
- x_{10} : Mix settlement index (coded from 1 if not mixed to 2 if mixed).

Example: Banknote Authentication

The bank dataset is designed to classify the status of a banknote. The status is represented as $y = 0$ for a genuine banknote and $y = 1$ for a false one. The classification is influenced by several dimensional features:

- x_1 : Length of the banknote.
- x_2 : Width of the left edge.
- x_3 : Width of the right edge.
- x_4 : Width of the bottom margin.

5.2 Regressors and Response Definition

To formally structure regression problems for a set of experimental subjects $i = 1, \dots, n$, we differentiate between predictors and outcomes:

Definition: Model Variables

- **Response Variable (y_i):** The variable of primary interest, called response or outcome variable. E.g., percent body fat, number of caterpillar colonies.
- **Explanatory Variables ($\bar{x}_i$):** Denoted as $\bar{x}_i = (x_{i1}, \dots, x_{ik})$, these are called covariates or explanatory variables, and they may be discrete, continuous or both.

The distribution of y_i given $\bar{x}_i$ is typically studied in the context of a set of units or experimental subjects, $i = 1, \dots, n$.

5.3 Goals of a Regression Model

Fundamentally, the objective is to find a proper representation of the distribution, $f(y|\theta, \bar{x})$, of an observable variable y given a vector of observables x , based on a sample of (x_i, y_i) pairs.

5.4 Machine Learning Point of View

From a machine learning perspective, the problem is formulated through several key concepts:

- **Input/Output Definition:** For an input x and output y , we model the probabilistic relation $y|\bar{x}, \theta \sim f(y|x, \theta)$.
- **Training Data:** We utilize a learning set represented by $(x_1, y_1), \dots, (x_n, y_n)$.
- **Target Parameter:** The parameter θ is the component to be learnt.
- **Inference:** Given a new observation $\bar{x}_{new}$, the goal is to predict the new outcome y_{new} via prediction/classification.

Important Note

While statistical models are excellent at finding associations, we must use them merely to understand how the x 's are related to the outcome y ; one must be careful as there is no causality implied!

5.5 Parameters and General Linear Models

In addition to y and x , one has the parameter θ . The most important part of θ is a vector $\beta = (\beta_0, \beta_1, \dots, \beta_k)$, meaning we can express the parameter space as $\theta = (\beta, \theta_{add})$. For simplicity in base linear models, we assume here that $\theta_{add} = \emptyset$ and let $\bar{x} = (1, \dots, x_k)$.

Theorem & Rules: General Linear Models (GLM)

In a General Linear Model, the mathematical relationship is generalized via a link function. Specifically, one has that:

- The conditional distribution is modeled as $f(y|\bar{x}, \beta) \sim f(y|\beta^T \bar{x})$ for a distribution f in the exponential family.
- The conditional expectation is defined as $\mathbb{E}[y|\bar{x}, \beta] = h(\beta^T \bar{x})$ for a suitable function h .
- The function h acts as the inverse of the so-called link function, connecting the linear predictor to the expected value of the response.

5.6 Normal Linear Models

In the ordinary normal linear regression framework, we assume a linear relationship between the covariates and the response variable. For each observation i , the model is defined as:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \epsilon_i = \beta^T \bar{x}_i + \epsilon_i$$

where the error terms ϵ_i are assumed to be independent and identically distributed (i.i.d.) normal random variables:

$$\epsilon_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$$

This can be explicitly written in matrix form to represent the entire system of equations for all n observations:

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

5.6.1 Multivariate Normal Distribution

Definition: Multivariate Gaussian Distribution

A random vector $y = (y_1, \dots, y_n)^t$ is said to follow a multivariate normal distribution with mean vector m and variance-covariance matrix Σ :

$$y \sim \mathcal{N}_n(m, \Sigma)$$

if its probability density function (defined on $\mathbb{R}^n$) is given by:

$$f(y|m, \Sigma) = \frac{1}{(2\pi)^{\frac{n}{2}} |\det(\Sigma)|} e^{-\frac{1}{2}(y-m)^t \Sigma^{-1} (y-m)}$$

Recall the fundamental properties of this distribution:

- The expected value of each component is $\mathbb{E}[y_i] = m_i$ for $i = 1, \dots, n$.
- The covariance between any two components is given by the matrix elements $Cov(y_i, y_j) = \Sigma_{ij}$.

5.6.2 Compact Notation and Vector Form

To write the model in a more compact and manageable matrix/vector form, we set the following notations:

- $y = (y_1, \dots, y_n)^t$ as the response vector.
- $\epsilon = (\epsilon_1, \dots, \epsilon_n)^t$ as the error vector.
- X as the $n \times (k+1)$ matrix of explanatory variables, which includes a leading column of ones for the intercept term.

Consequently, the Normal Linear Model can be succinctly expressed as:

$$y = X\beta + \epsilon$$

In particular, conditional on the parameters and the design matrix, the distribution of the response vector is a multivariate normal:

$$y|\beta, \sigma^2, X \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$$

This structure implies the following conditional moments:

- The conditional expectation is $\mathbb{E}[y_i|\beta, X] = \beta^T \bar{x}_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik}$.
- The conditional variance is constant across observations, $Var(y_i|\sigma^2, X) = \sigma^2$.
- The observations are conditionally independent, meaning $Cov(y_i, y_j) = 0$ for $i \neq j$.

Important Note

Note that this formulation is perfectly aligned with the General Linear Model (GLM) framework where the link function is the identity function ($h(z) = z$) and the additional parameter is the variance ($\theta_{add} = \sigma^2$).

5.7 Objectives of Inference in Linear Models

When working with linear models, the typical objects of statistical inference include several key tasks:

- **Parameter Estimation:** Finding robust estimates for the unknown parameters, specifically the regression coefficients β_i and the standard deviation σ .
- **Interval Estimation:** Constructing confidence or credible intervals for the parameters to quantify estimate uncertainty.
- **Prediction:** Given a new set of covariates $x_{new} = (x_{new,1}, \dots, x_{new,k})$, the goal is to find a predictive distribution for the new response $y_{new}|x_{new}$.
- **Hypothesis Testing:** Performing statistical tests, for example, checking if a specific covariate has a significant effect ($\beta_i = 0$ vs $\beta_i \neq 0$ for a given i).
- **Model Selection:** Determining which covariates need to be included in the model to balance predictive power and complexity.

5.8 Frequentist Estimation and Likelihood (Optional)

Before delving into the Bayesian framework, it is instructive to review the classical frequentist approach to linear models for the sake of comparison.

5.8.1 Likelihood Function and Maximum Likelihood Estimator

The likelihood function for the ordinary normal linear model represents the joint probability density of the observed data given the parameters. It is expressed as:

$$f(y|X, \beta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} (y - X\beta)^t (y - X\beta) \right]$$

The maximum likelihood estimator (MLE) of the coefficient vector β is found by maximizing this likelihood, which is mathematically equivalent to solving the least squares minimization problem:

$$\min_{\beta} (y - X\beta)^t (y - X\beta) = \min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_k x_{ik})^2$$

Solving this optimization problem yields the well-known closed-form solution for the estimator:

$$\hat{\beta} = (X^t X)^{-1} X^t y$$

Important Note

Convention: For this estimator to be uniquely defined, we must assume that the design matrix X is of full rank. Equivalently, this means that the square matrix $X^t X$ is invertible.

5.8.2 Properties of the Least Squares Estimator

The estimator $\hat{\beta}$ possesses several important statistical properties:

- **Unbiasedness:** It is an unbiased estimator of the true parameter β , meaning that on average, it hits the true value: $\mathbb{E}[\hat{\beta}|\beta] = \beta$.
- **Variance:** The conditional variance of the estimator is given by $Var(\hat{\beta}|\sigma^2, X) = \sigma^2(X^t X)^{-1}$.

Additionally, we need to estimate the error variance σ^2 . The standard unbiased estimator for σ^2 relies on the residual sum of squares:

$$\hat{\sigma}^2 = \frac{1}{n - k - 1} (y - X\hat{\beta})^t (y - X\hat{\beta}) = \frac{s^2}{n - k - 1}$$

5.8.3 Frequentist Linear Models using R

To estimate a linear model using a frequentist approach in practice, one can use the built-in R function for linear modeling:

Example: R Implementation

The standard command in R is `lm()`:

```
pinco = lm(y ~ X)
summary(pinco)
```

Using the `summary()` function will display the parameter estimates, their standard errors, and other basic diagnostic information.

Important Note

Design Matrix Note: When using the `lm(y ~ X)` syntax, R automatically adds the column of ones (the intercept term) to the design matrix. Therefore, the matrix X you supply to the formula should *not* explicitly include the column of ones.

6 Lesson 6: Bayesian Linear Models and Priors

Returning to the Bayesian framework, the normal linear model consists of a defined likelihood function and a prior distribution defined over the parameters.

6.1 Likelihood and Prior Specification

The general Bayesian linear model is specified as follows:

- **Likelihood:** $y|\beta, \sigma^2, X \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$
- **Prior:** $(\beta, \sigma^2) \sim \pi(\beta, \sigma^2)$

Depending on the inferential goals and available information, various prior distributions can be chosen:

- Conjugate prior
- Zellner's (improper) informative G-prior
- Mixture of G-prior
- Lasso
- Many others (e.g., spike and slab, hierarchical)

6.2 Alternative Parametrization

There is another common parametrization of the linear regression problem, which is slightly different from the standard matrix formulation. In this setup, the intercept β_0 is separated from the other coefficients:

$$y|\beta, \sigma^2, X \sim \mathcal{N}_n(\beta_0 \mathbf{1} + \tilde{X}\beta, \sigma^2 I_n)$$
$$(\beta_0, \beta, \sigma^2) \sim \pi(\beta_0, \beta, \sigma^2)$$

Important Note

In this formulation, $\beta = (\beta_1, \dots, \beta_k)$ represents only the slope coefficients. The matrix $\tilde{X}$ is the same design matrix X but *without* the column of ones. We shall use this parametrization when using the BAS library in R!

6.3 Overview of Common Priors

Definition: Conjugate Prior

The standard conjugate prior for the normal linear model relies on a Normal-Inverse-Gamma structure:

- $\beta|\sigma^2 \sim \mathcal{N}_{k+1}(\tilde{\beta}, \sigma^2 M)$
- $\sigma^2 \sim \mathcal{IG}(a, b)$

Here, the hyper-parameters are $\tilde{\beta}$, $a > 0$, $b > 0$, and a positive-definite matrix $M > 0$.

Definition: Zellner's Informative G-Prior

Zellner's G-prior (as described in the book) simplifies the covariance matrix specification by linking it to the design matrix:

- $\beta|\sigma^2 \sim \mathcal{N}_{k+1}(\tilde{\beta}, c\sigma^2(X^t X)^{-1})$
- $\sigma^2 \sim \pi(\sigma^2) = \sigma^{-2}$ (an improper prior)

In the alternative parametrization (used in BAS), setting $\tilde{\beta} = 0$ and $\alpha = c$, it is expressed as:

- $\beta = (\beta_1, \dots, \beta_k)|\sigma^2 \sim \mathcal{N}_k(0, \alpha\sigma^2(\tilde{X}^t \tilde{X})^{-1})$
- $(\beta_0, \sigma^2) \sim \pi(\beta_0, \sigma^2) = \sigma^{-2}$

Definition: Hierarchical and Mixture Priors

Often, simple priors do not yield closed-form posteriors, so a hierarchical structure is used.

Simple Hierarchical:

- $\beta_j|\sigma^2, \sigma_\beta^2 \sim \mathcal{N}(0, \sigma_\beta^2)$
- $\sigma_\beta^2 \sim \mathcal{IG}(a_\beta, b_\beta)$
- $\sigma^2 \sim \mathcal{IG}(a_\sigma, b_\sigma)$

Zellner-Siow Prior (Mixture of G-prior):

- $\beta = (\beta_1, \dots, \beta_k)|\sigma^2, \alpha, X \sim \mathcal{N}_k(0, \alpha\sigma^2(X^t X)^{-1})$
- $\sigma^2|X, \alpha \sim \pi(\sigma^2) = \sigma^{-2}$
- $1/\alpha \sim \pi_0 = \text{Gamma}(1/2, n/2)$

Definition: Priors for Variable Selection

A typical drawback in regression problems is the fact that one has too many covariates. We can combat this using specific priors: **Bayesian Lasso**:

- $\beta_j | \sigma^2, \sigma_\beta^2 \sim \mathcal{DE}(\sigma_\beta^2)$ (Double Exponential)
- $\sigma_\beta^2 \sim \mathcal{IG}(a_\beta, b_\beta)$
- $\sigma^2 \sim \mathcal{IG}(a_\sigma, b_\sigma)$

Bayesian Spike and Slab:

- $(\beta_0, \sigma^2) \sim \pi(\beta_0, \sigma^2) = \sigma^{-2}$
- $\beta_j | \delta_j, \rho \sim \mathcal{N}(0, \sigma_\beta^2) \mathbb{I}\{\delta_j = 1\} + \delta_0(\beta_j) \mathbb{I}\{\delta_j = 0\}$
- $\delta_j | \rho \sim \text{Be}(\rho)$ and $\rho \sim \text{Beta}(a, b)$

The latent variable δ_j selects the component β_j that is set equal to 0. If $\delta_j = 0$, then $\beta_j = 0$; if $\delta_j = 1$, then β_j is sampled from a Gaussian with $\sigma_\beta^2 \gg 0$.

6.4 Conjugate Prior Posterior Updating

Theorem & Rules: Posterior Distribution for Conjugate Prior

Assume the conjugate prior for the Linear Model:

$$\beta | \sigma^2, X \sim \mathcal{N}_{k+1}(\tilde{\beta}, \sigma^2 M^{-1}), \quad \sigma^2 | X \sim \mathcal{IG}(a, b)$$

where M is a $(k+1) \times (k+1)$ positive definite symmetric matrix.

In this case, the posterior distribution has the exact same distributional form as the prior, with updated parameters:

- $M_{\text{updated}} = (M + X^t X)^{-1}$
- $\tilde{\beta}_{\text{updated}} = (M + X^t X)^{-1} \{ (X^t X) \hat{\beta} + M \tilde{\beta} \}$
- $a_{\text{updated}} = \frac{n}{2} + a$
- $b_{\text{updated}} = b + \frac{s^2}{2} + \frac{(\tilde{\beta} - \hat{\beta})^t (M^{-1} + (X^t X)^{-1})^{-1} (\tilde{\beta} - \hat{\beta})}{2}$

Recall that $\hat{\beta}$ is the Maximum Likelihood Estimator (MLE) of β .

6.5 The Experimenter's Dilemma and Prior Selection

In Bayesian linear regression, a significant challenge arises when setting the hyper-parameters for the prior distributions, commonly referred to as the experimenter's dilemma. Specifically, one must

choose the precision matrix M or the scalar c if we assume a structure like $M = I_{k+1}/c$.

Example: Processionary Caterpillar Prior Sensitivity

Consider a scenario where we have no precise prior information regarding the parameters $\tilde{\beta}$, M , a , and b . We might set $a = 2.1$ and $b = 2$, which yields a prior mean for σ^2 of 1.82 and a prior variance of 33.06, alongside a zero-vector for the prior mean of the coefficients, $\tilde{\beta} = 0_{k+1}$.

The choice of the scaling factor c exerts a lasting influence on the posterior estimates. The table below demonstrates how varying c impacts the posterior expectations and variance for the intercept β_0 and error variance σ^2 :

c	$\mathbb{E}^\pi(\sigma^2 y, X)$	$\mathbb{E}^\pi(\beta_0 y, X)$	$Var^\pi(\beta_0 y, X)$
0.1	1.0044	0.1251	0.0988
1	0.8541	0.9031	0.7733
10	0.6976	4.7299	3.8991
100	0.5746	9.6626	6.8355
1000	0.5470	10.8476	7.3419

The experimental prior determination is effectively restricted to the choices of $\tilde{\beta}$ and the constant c . The factor c can be interpreted as a measure of the amount of information available in the prior relative to the sample. For instance, setting $1/c = 0.5$ (or $c = 2$) gives the prior the same weight as 50% of the sample. As seen in the example, there is a lasting influence of this factor c on the final inference.

6.6 Zellner's Informative G-Prior in Detail

Definition: Zellner's G-Prior

Zellner's G-prior introduces a constraint that allows the experimenter to input information about the location parameter of the regression ($\tilde{\beta}$) while bypassing the most difficult aspects of prior specification, namely the derivation of the prior correlation structure.

Zellner's (improper) prior relies on the design matrix X to scale the covariance and is defined as follows:

$$\beta|\sigma^2, X \sim \mathcal{N}_{k+1}(\tilde{\beta}, c\sigma^2(X^t X)^{-1})$$

$$\sigma^2|X \sim \pi(\sigma^2) = \sigma^{-2}$$

The essential hyper-parameters strictly required from the user are $c > 0$ and $\tilde{\beta}$.

6.6.1 Posterior Structure and Proof Outline

Theorem & Rules: Posterior under Zellner's G-Prior

Under Zellner's improper prior, the conditional posterior distributions for the parameters follow closed-form known distributions:

The conditional posterior of the coefficients is Gaussian:

$$\beta | \sigma^2, y, X \sim \mathcal{N}_{k+1} \left(\frac{c}{c+1} \left(\frac{\tilde{\beta}}{c} + \hat{\beta} \right), \frac{\sigma^2 c}{c+1} (X^t X)^{-1} \right)$$

The marginal posterior of the variance follows an Inverse-Gamma distribution:

$$\sigma^2 | y, X \sim \mathcal{IG} \left(\frac{n}{2}, \frac{s^2}{2} + \frac{1}{2(c+1)} (\tilde{\beta} - \hat{\beta})^t X^t X (\tilde{\beta} - \hat{\beta}) \right)$$

By integrating out σ^2 , the marginal posterior for the coefficients β follows a multivariate Student's t-distribution:

$$\beta | y, X \sim \mathcal{T}_{k+1} \left(n, \frac{c}{c+1} \left(\frac{\tilde{\beta}}{c} + \hat{\beta} \right), \frac{c(s^2 + (\tilde{\beta} - \hat{\beta})^t X^t X (\tilde{\beta} - \hat{\beta}) / (c+1))}{n(c+1)} (X^t X)^{-1} \right)$$

The proof for this posterior structure relies on combining the likelihood and the prior:

$$\pi(\beta, \sigma^2 | y, X) \propto f(y | \beta, \sigma^2, X) \pi(\beta, \sigma^2 | X)$$

Because the term $X^t X$ is utilized in both the prior covariance and the likelihood, we can apply a mathematical technique often referred to as the "G-prior trick". This trick leverages the decomposition of the sum of squared errors:

$$(y - X\beta)^t (y - X\beta) = (y - X\hat{\beta})^t (y - X\hat{\beta}) + (\beta - \hat{\beta})^t X^t X (\beta - \hat{\beta})$$

This identity follows directly from the orthogonality condition $(X\hat{\beta})^T X\beta = y^T X\beta$, which is a consequence of the expression for the MLE $\hat{\beta}$. By substituting this into the likelihood and combining it with the Gaussian prior term $\exp \left[-\frac{1}{2c\sigma^2} (\beta - \tilde{\beta})^t X^t X (\beta - \tilde{\beta}) \right]$, we can complete the square to derive the updated normal parameters.

6.6.2 Bayes Estimators and Conjugacy

The Bayes estimators (under squared error loss) for β and σ^2 are simply the expectations of their respective posterior distributions.

For the coefficient vector, the estimator is a weighted average of the prior mean and the Maximum Likelihood Estimator:

$$\mathbb{E}^\pi[\beta | y, X] = \frac{1}{c+1} (\tilde{\beta} + c\hat{\beta})$$

For the variance term, the estimator is:

$$\mathbb{E}^\pi[\sigma^2|y, X] = \frac{s^2 + (\tilde{\beta} - \hat{\beta})^t X^t X (\tilde{\beta} - \hat{\beta}) / (c + 1)}{n - 2}$$

Important Note

The property of conjugacy acts as a convenient tool for translating prior information regarding β .

- If we set $c = 1$, this is mathematically equivalent to placing the exact same weight on our prior information as we do on the data sample. The estimator becomes a simple average: $\mathbb{E}^\pi(\beta|y, X) = \frac{\tilde{\beta} + \hat{\beta}}{2}$.
- Conversely, if $c = 100$, the prior receives a relative weight of only 1% compared to the sample.
- Note that it is only when c goes to infinity that the influence of the prior truly vanishes, effectively recovering the frequentist estimates.

Example: Pine Processionary Caterpillars: Posterior Estimates

To illustrate the lasting influence of the prior scaling factor c , we can examine the posterior expectations and variances for the caterpillar dataset under two different configurations.

When $c = 100$, the posterior moments for the coefficients are:

Coefficient	$\mathbb{E}^\pi(\beta_i y, X)$	$\text{Var}^\pi(\beta_i y, X)$	Coefficient	$\mathbb{E}^\pi(\beta_i y, X)$	$\text{Var}^\pi(\beta_i y, X)$
β_0	10.8895	6.4094	β_6	-0.3532	1.6793
β_1	-0.0044	2e-06	β_7	-0.2351	0.6926
β_2	-0.0533	0.0003	β_8	0.1793	0.0383
β_3	0.0673	0.0068	β_9	-1.2726	0.5119
β_4	-1.2808	0.2175	β_{10}	-0.4288	0.3696
β_5	0.2293	0.0075			

When the prior weight is further diluted by setting $c = 1,000$, the estimates shift slightly but remain largely consistent:

Coefficient	$\mathbb{E}^\pi(\beta_i y, X)$	$\text{Var}^\pi(\beta_i y, X)$	Coefficient	$\mathbb{E}^\pi(\beta_i y, X)$	$\text{Var}^\pi(\beta_i y, X)$
β_0	10.9874	6.2604	β_6	-0.3564	1.6403
β_1	-0.0044	2e-06	β_7	-0.2372	0.6765
β_2	-0.0538	0.0003	β_8	0.1809	0.0375
β_3	0.0679	0.0066	β_9	-1.2840	0.5100
β_4	-1.2923	0.2125	β_{10}	-0.4327	0.3670
β_5	0.2314	0.0073			

This demonstrates that as c increases, the estimates stabilize, increasingly dominated by the likelihood of the observed data.

6.7 Prediction in Bayesian Linear Models

A fundamental goal in regression is the prediction of $m \geq 1$ future observations originating from new units. For these units, the explanatory variables, denoted as the matrix X_{new} , have been observed, but the outcome variable y_{new} remains unknown.

The *prior predictive distribution* for these new observations is a multivariate normal:

$$y_{\text{new}}|\sigma^2, \beta, X_{\text{new}} \sim \mathcal{N}_m(X_{\text{new}}\beta, \sigma^2 I_m)$$

However, for robust inference, we must incorporate the information learned from our training data. The *posterior predictive distribution* integrates over the uncertainty in the parameters β and σ^2 . It is analytically derived by computing the following integral:

$$\int \pi(y_{\text{new}}|\sigma^2, \beta, X_{\text{new}})\pi(\sigma^2, \beta|y, X)d\sigma^2 d\beta$$

This integral admits an analytic expression, which can be referenced in the main textbook.

Theorem & Rules: Bayesian Predictor

Under a squared error loss function, the optimal point predictor is the mean of the posterior predictive distribution. It is given by:

$$X_{new} \frac{\tilde{\beta} + c\hat{\beta}}{c+1}$$

This representation is highly intuitive, as it is simply the matrix product of the new explanatory variables X_{new} and the Bayes estimate of the coefficient vector β (where $\hat{\beta}$ is the frequentist Maximum Likelihood Estimator).

6.8 Credible Regions and HPD Intervals (Optional)

To quantify the uncertainty of our estimates, we compute Highest Posterior Density (HPD) regions on subvectors of the parameter β . These are derived directly from the marginal posterior distribution.

For a single parameter β_i , the marginal distribution is a Student's t-distribution:

$$\beta_i|y, X \sim \mathcal{T} \left(n, \frac{c}{c+1} \left(\frac{\tilde{\beta}_i}{c} + \hat{\beta}_i \right), \frac{c(s^2 + (\tilde{\beta} - \hat{\beta})^t X^t X (\tilde{\beta} - \hat{\beta}) / (c+1))}{n(c+1)} \omega_{(i,i)} \right)$$

where $\omega_{(i,i)}$ corresponds to the (i, i) -th element of the matrix $(X^t X)^{-1}$.

To construct the interval, we define the location parameter $\tau = \frac{\tilde{\beta} + c\hat{\beta}}{c+1}$ and the scale matrix $K = \frac{c(s^2 + (\tilde{\beta} - \hat{\beta})^t X^t X (\tilde{\beta} - \hat{\beta}) / (c+1))}{n(c+1)} (X^t X)^{-1}$, letting its elements be denoted as $\kappa_{(i,j)}$. Using the standard normalizer transform:

$$\mathfrak{Z}_i = \frac{\beta_i - \tau_i}{\sqrt{\kappa_{(i,i)}}}$$

which follows a standard t-distribution with n degrees of freedom, a $1 - \alpha$ HPD interval for β_i is expressed as:

$$\left[\tau_i - \sqrt{\kappa_{(i,i)}} F_n^{-1}(1 - \alpha/2), \tau_i + \sqrt{\kappa_{(i,i)}} F_n^{-1}(1 - \alpha/2) \right]$$

Example: HPD Intervals for Caterpillar Dataset

Setting $c = 100$, the specific HPD intervals for the covariates in the processionary caterpillar dataset reflect the uncertainty in their respective influences:

- β_0 : $[5.7435, 16.2533]$
- β_1 : $[-0.0071, -0.0018]$
- ...
- β_4 (height of central tree): $[-2.2618, -0.3255]$
- β_5 (diameter of central tree): $[0.0524, 0.4109]$

Intervals that do not contain zero (such as β_1 and β_4) suggest a statistically credible linear influence of those variables on the outcome.

6.9 Handling Categorical Variables

When structuring the design matrix X , a critical distinction exists between finite-valued numerical regressors (like the orientation of the area x_7 in the caterpillar dataset) and pure categorical variables (or factors). Categorical variables also take a finite number of values, but their range holds no inherent numerical meaning.

Example: Socio-Professional Category

If an explanatory variable x represents the socio-professional category of an employee, it might range from 1 to 9 for a rough grid of activities, and from 1 to 89 on a finer grid. However, these numerical values are not comparable (e.g., category 4 is not "twice" category 2).

Because it makes little sense to involve such a variable directly in the regression equation, we must replace the single categorical regressor x (assuming $x \in \{1, \dots, m\}$) with m distinct indicator (or dummy) variables:

$$x_1 := \mathbb{I}_1(x), \dots, x_m := \mathbb{I}_m(x)$$

This allows for the use of a different constant coefficient for each class value of the categorical variable:

$$\mathbb{E}[y_i | \beta, X] = \dots + \beta_1 \mathbb{I}_1(x) + \dots + \beta_m \mathbb{I}_m(x) + \dots$$

Important Note

To prevent perfect multicollinearity and satisfy the full rank condition on the matrix X , one must eliminate one class from the dummy variables when building the model. The dropped class acts as the baseline reference category.

6.10 Practical Implementation in R

For applied Bayesian modeling and variable selection, practical examples frequently leverage the BAS (Bayesian Adaptive Sampling) library in R. Specific implementations and code structures are often explored interactively in supplementary R-Notebooks (e.g., `NotebookLM.BAS`).

6.11 Advanced Topics: Testing and Model Selection (Optional)

For an in-depth theoretical treatment of Hypothesis Testing and Model Selection in Bayesian linear regression, one should refer to Chapter 3 of the textbook "Bayesian Core" by C. Robert and J.M. Marin. While the following advanced concepts are considered optional for the main theoretical progression, they provide a crucial foundation for the applied techniques that will be explored in the Laboratory sessions. Reviewing this material is highly recommended to fully understand the computational tools used in practice.

6.11.1 Marginal Distribution of the Response

The marginal distribution of the response variable y can be derived directly from the joint model specifications.

Theorem & Rules: Marginal Distribution of y

Under Zellner's G-prior, the coefficient vector follows a normal distribution:

$\beta|\sigma^2, X \sim \mathcal{N}_{k+1}(\tilde{\beta}, c\sigma^2(X^t X)^{-1})$. Consequently, the linear predictor satisfies $X\beta|\sigma^2, X \sim \mathcal{N}(X\tilde{\beta}, c\sigma^2 X(X^t X)^{-1} X^t)$.

This implies that the conditional distribution of y , given only the variance and the design matrix, is:

$$y|\sigma^2, X \sim \mathcal{N}_n(X\tilde{\beta}, \sigma^2(I_n + cX(X^t X)^{-1} X^t))$$

By analytically integrating out the variance parameter σ^2 , we obtain the marginal distribution of y , which takes the form of a multivariate Student's t-distribution:

$$\begin{aligned} f(y|X) &= (c+1)^{-(k+1)/2} \pi^{-n/2} \Gamma(n/2) \\ &\times \left[y^t y - \frac{c}{c+1} y^t X (X^t X)^{-1} X^t y \right. \\ &\quad \left. - \frac{1}{c+1} \tilde{\beta}^t X^t X \tilde{\beta} \right]^{-n/2} \end{aligned}$$

6.11.2 Point Null Hypothesis Testing

Definition: Point Null Hypothesis

If a point null hypothesis takes the form $H_0 : R\beta = r$, the regression model under H_0 can be rewritten using a restricted parameter space. Let β^0 be the constrained $(k+1-q)$ -dimensional parameter vector. The model under the null hypothesis becomes:

$$y|\beta^0, \sigma^2, X_0 \sim \mathcal{N}_n(X_0\beta^0, \sigma^2 I_n)$$

Under the prior for this restricted model, $\beta^0|X_0, \sigma^2 \sim \mathcal{N}_{k+1-q}(\tilde{\beta}^0, c_0\sigma^2(X_0^t X_0)^{-1})$. Following the same integration steps as the full model, the marginal distribution of y under H_0 is computed as:

$$\begin{aligned} f(y|X_0, H_0) &= (c_0 + 1)^{-(k+1-q)/2} \pi^{-n/2} \Gamma(n/2) \\ &\times \left[y^t y - \frac{c_0}{c_0 + 1} y^t X_0 (X_0^t X_0)^{-1} X_0^t y \right. \\ &\quad \left. - \frac{1}{c_0 + 1} (\tilde{\beta}_0)^t X_0^t X_0 \tilde{\beta}_0 \right]^{-n/2} \end{aligned}$$

Theorem & Rules: Bayes Factor

The Bayes factor, which quantifies the evidence for the alternative hypothesis H_1 relative to H_0 , has a closed-form solution due to the conjugacy of the G-prior:

$$B_{10}^\pi = \frac{f(y|X, H_1)}{f(y|X_0, H_0)}$$

Substituting the respective marginals, we get:

$$\begin{aligned} B_{10}^\pi &= \frac{(c_0 + 1)^{(k+1-q)/2}}{(c + 1)^{(k+1)/2}} \\ &\times \left[\frac{y^t y - \frac{c_0}{c_0+1} y^t X_0 (X_0^t X_0)^{-1} X_0^t y - \frac{1}{c_0+1} (\tilde{\beta}_0)^t X_0^t X_0 \tilde{\beta}_0}{y^t y - \frac{c}{c+1} y^t X (X^t X)^{-1} X^t y - \frac{1}{c+1} \tilde{\beta}^t X^t X \tilde{\beta}} \right]^{n/2} \end{aligned}$$

Important Note

When utilizing the Bayes Factor in this formulation, two analytical caveats must be kept in mind:

- The calculation inherently means using the same error variance σ^2 across both competing models.
- The final value still heavily depends on the arbitrary choice of the prior scaling constants (c_0, c) .

6.11.3 Variable Selection

Returning to the fundamental regression setup, we typically observe one dependent random variable y and a set $\{x_1, \dots, x_k\}$ of k potential explanatory variables. A central question naturally arises: are all of these X_i variables truly involved in the regression equation?

To investigate this systematically, we make a core structural assumption: every possible subset $\{i_1, \dots, i_q\}$ containing q explanatory variables (where $0 \leq q \leq k$), denoted as $\{1_n, x_{i_1}, \dots, x_{i_q}\}$, constitutes a proper set of explanatory variables for the regression of y . Importantly, the intercept term is always included in every corresponding model evaluated.

Example: Subsets of Variables

Suppose we have a full model with an intercept and three predictors, resulting in the coefficient vector $\beta = (\beta_0, \beta_1, \beta_2, \beta_3)$. Since the intercept β_0 is always included as a baseline, the possible active predictor subsets are:

- **All predictors:** (1, 2, 3)
- **Two predictors:** (1, 2), (1, 3), (2, 3)
- **One predictor:** (1), (2), (3)
- **Null model:** Only the intercept.

Important Note

Computational Issue: Evaluating all combinations means there are exactly 2^k models in competition. As the number of candidate regressors k grows, calculating the posterior for every single combination rapidly becomes computationally intractable.

6.11.4 Model Notations and Indicators

To systematically address the variable selection problem, we must formally define the mathematical space of potential models. Let the full design matrix, containing the column of ones 1_n and all k potential predictor variables, be denoted as $X = \begin{bmatrix} 1_n & x_1 & \cdots & x_k \end{bmatrix}$.

Definition: Model Indicator Vector

Each candidate model $\mathfrak{M}_\gamma$ is uniquely associated with a binary indicator vector $\gamma \in \Gamma = \{0, 1\}^k$. The condition $\gamma_i = 1$ signifies that the explanatory variable x_i is actively included in the model $\mathfrak{M}_\gamma$.

From this vector, we derive several important quantities:

- $q_\gamma = 1_n^t \gamma$: The total number of explanatory variables included in the model.
- $t_1(\gamma)$ and $t_0(\gamma)$: The sets of indices corresponding to the variables that are included and excluded from the model, respectively.

Based on the active indices, we can partition our parameters and data for any given model $\mathfrak{M}_\gamma$. For the full parameter vector $\beta \in \mathbb{R}^{k+1}$, we define β_γ as the specific subvector containing the intercept and the active coefficients: $\beta_\gamma = (\beta_0, (\beta_i)_{i \in t_1(\gamma)})$. Similarly, X_γ is defined as the submatrix of X formed by retaining only the column 1_n and the predictor columns indexed by $t_1(\gamma)$.

The specific model $\mathfrak{M}_\gamma$ in competition is thus formally defined by the conditional distribution:

$$y|\gamma, \beta_\gamma, \sigma^2, X \sim \mathcal{N}_n(X_\gamma \beta_\gamma, \sigma^2 I_n)$$

where $\beta_\gamma \in \mathbb{R}^{q_\gamma+1}$ and the variance $\sigma^2 \in \mathbb{R}_+^*$ represent the unknown parameters of this chosen submodel.

Important Note

Variance Assumption: The error variance parameter σ^2 is common to all models in the selection space. Thus, we utilize the exact same prior formulation for σ^2 across all candidate models.

6.11.5 Priors for Model Competition

When faced with 2^k competing models, it is computationally and practically unreasonable to expect a practitioner to specify a prior for every single $\mathfrak{M}_\gamma$ in a completely subjective and autonomous manner.

Theorem & Rules: Informative G-Prior Shortcut

To circumvent this issue, we employ a shortcut: we derive all model-specific priors from a single global prior associated with the "full model", which corresponds to the vector $\gamma = (1, \dots, 1)$.

For the full model, Zellner's G-prior is assumed:

$$\beta|\sigma^2, X \sim \mathcal{N}_{k+1}(\tilde{\beta}, c\sigma^2(X^t X)^{-1}) \quad \text{and} \quad \sigma^2 \sim \pi(\sigma^2|X) = \sigma^{-2}$$

By projection, for each submodel $\mathfrak{M}_\gamma$, the prior distribution of β_γ conditional on the variance is naturally fixed as:

$$\beta_\gamma|\gamma, \sigma^2 \sim \mathcal{N}_{q_\gamma+1}(\tilde{\beta}_\gamma, c\sigma^2(X_\gamma^t X_\gamma)^{-1})$$

where the projected mean is $\tilde{\beta}_\gamma = (X_\gamma^T X_\gamma)^{-1} X_\gamma^T \tilde{\beta}$.

The resulting joint prior for the model $\mathfrak{M}_\gamma$ is an improper prior given by:

$$\pi(\beta_\gamma, \sigma^2|\gamma) \propto (\sigma^2)^{-(q_\gamma+1)/2-1} \exp \left[-\frac{1}{2(c\sigma^2)} (\beta_\gamma - \tilde{\beta}_\gamma)^t (X_\gamma^t X_\gamma) (\beta_\gamma - \tilde{\beta}_\gamma) \right]$$

6.11.6 Posterior Model Probabilities

To navigate the space of models, there are infinitely many ways of defining a prior on the model index γ . A standard, non-informative choice is the uniform prior across the model space: $\pi(\gamma|X) = 2^{-k}$.

The posterior distribution of the model indicator is central to Bayesian variable selection, as it is strictly proportional to the marginal density of y under $\mathfrak{M}_\gamma$, which is also known as the evidence of $\mathfrak{M}_\gamma$. Mathematically, this is expressed as:

$$\pi(\gamma|y, X) \propto f(y|\gamma, X)\pi(\gamma|X) \propto f(y|\gamma, X)$$

The evidence is computed by integrating out both β and σ^2 :

$$f(y|\gamma, X) = \int \left(\int f(y|\gamma, \beta, \sigma^2, X) \pi(\beta|\gamma, \sigma^2, X) d\beta \right) \pi(\sigma^2|X) d\sigma^2$$

Evaluating the inner integral yields the conditional marginal likelihood:

$$f(y|\gamma, \sigma^2, X) = \frac{(c+1)^{-(q_\gamma+1)/2}}{(2\pi\sigma^2)^{n/2}} \exp \left(-\frac{1}{2\sigma^2} y^t y + \frac{c y^t X_\gamma (X_\gamma^t X_\gamma)^{-1} X_\gamma^t y - \tilde{\beta}_\gamma^t X_\gamma^t X_\gamma \tilde{\beta}_\gamma}{2\sigma^2(c+1)} \right)$$

Integrating this expression with respect to the variance σ^2 provides the exact, closed-form posterior probability of the model structure:

$$\pi(\gamma|y, X) \propto (c+1)^{-(q_\gamma+1)/2} \left[y^t y - \frac{c}{c+1} y^t X_\gamma (X_\gamma^t X_\gamma)^{-1} X_\gamma^t y - \frac{1}{c+1} \tilde{\beta}_\gamma^t X_\gamma^t X_\gamma \tilde{\beta}_\gamma \right]^{-n/2}$$

Example: Variable Selection on Caterpillars Dataset

Applying this variable selection approach to the Pine processionary caterpillars dataset reveals the most probable combinations of covariates.

The top performing models ranked by their posterior probability $\pi(\gamma|y, X)$ are:

Included Variables: $t_1(\gamma)$	Posterior Prob: $\pi(\gamma y, X)$
0, 1, 2, 4, 5	0.2316
0, 1, 2, 4, 5, 9	0.0374
0, 1, 9	0.0344
0, 1, 2, 4, 5, 10	0.0328
0, 1, 4, 5	0.0306

Interpretation: The model $\mathfrak{M}_\gamma$ that achieves the highest posterior probability (0.2316) corresponds to the subset $t_1(\gamma) = (1, 2, 4, 5)$ (with the intercept, 0, implicitly included). This optimal subset successfully isolates the most influential environmental predictors:

- **Altitude** (x_1)
- **Slope** (x_2)
- **Height of the central tree** (x_4)
- **Diameter of the central tree** (x_5)

This precise subset accurately corresponds to the five variables identified as significant in the standard frequentist R regression output.